

| Tool Version : Vivado v.2025.2 (win64) Build 6299465 Fri Nov 14
19:35:11 GMT 2025
| Date : Wed Jul 1 22:38:49 2026
| Host : AYODELE running 64-bit major release (build 9200)
| Command : report_timing_summary -max_paths 10 -routable_nets
-report_unconstrained -file
Matrix_Accelerator_timing_summary_routed.rpt -pb
Matrix_Accelerator_timing_summary_routed.pb -rpx
Matrix_Accelerator_timing_summary_routed.rpx -warn_on_violation
| Design : Matrix_Accelerator
| Device : 7k70t-fbv676
| Speed File : -1 PRODUCTION 1.12 2017-02-17
Design State : Routed

Timing Summary Report

Timer Settings

Enable Multi Corner Analysis	:	Yes
Enable Pessimism Removal	:	Yes
Pessimism Removal Resolution	:	Nearest Common Node
Enable Input Delay Default Clock	:	No

Enable Preset / Clear Arcs	:	No
Disable Flight Delays	:	No
Ignore I/O Paths	:	No
Timing Early Launch at Borrowing Latches	:	No
Borrow Time for Max Delay Exceptions	:	Yes
Merge Timing Exceptions	:	Yes
Inter-SLR Compensation	:	Conservative

Corner	Analyze	Analyze
Name	Max Paths	Min Paths
-----	-----	-----
Slow	Yes	Yes
Fast	Yes	Yes

Report Methodology

Rule	Severity	Description	Violations
-----	-----	-----	-----
TIMING-18	Warning	Missing input or output delay	153

Note: This report is based on the most recent report_methodology run and may not be up-to-date. Run report_methodology on the current design for the latest report.

check_timing report

Table of Contents

1. checking no_clock (0)

```
2. checking constant_clock (0)
3. checking pulse_width_clock (0)
4. checking unconstrained_internal_endpoints (0)
5. checking no_input_delay (24)
6. checking no_output_delay (129)
7. checking multiple_clock (0)
8. checking generated_clocks (0)
9. checking loops (0)
10. checking partial_input_delay (0)
11. checking partial_output_delay (0)
12. checking latch_loops (0)
```

```
1. checking no_clock (0)
```

```
-----
```

```
There are 0 register/latch pins with no clock.
```

```
2. checking constant_clock (0)
```

```
-----
```

```
There are 0 register/latch pins with constant_clock.
```

```
3. checking pulse_width_clock (0)
```

```
-----
```

```
There are 0 register/latch pins which need pulse_width check
```

```
4. checking unconstrained_internal_endpoints (0)
```

```
-----
```

```
There are 0 pins that are not constrained for maximum delay.
```

```
There are 0 pins that are not constrained for maximum delay due to
constant clock.
```

```
5. checking no_input_delay (24)
```

```
-----
```

There are 24 input ports with no input delay specified. (HIGH)

There are 0 input ports with no input delay but user has a false path constraint.

6. checking no_output_delay (129)

There are 129 ports with no output delay specified. (HIGH)

There are 0 ports with no output delay but user has a false path constraint

There are 0 ports with no output delay but with a timing clock defined on it or propagating through it

7. checking multiple_clock (0)

There are 0 register/latch pins with multiple clocks.

8. checking generated_clocks (0)

There are 0 generated clocks that are not connected to a clock source.

9. checking loops (0)

There are 0 combinational loops in the design.

10. checking partial_input_delay (0)

There are 0 input ports with partial input delay specified.

11. checking partial_output_delay (0)

There are 0 ports with partial output delay specified.

12. checking latch_loops (0)

There are 0 combinational latch loops in the design through latch input

Design Timing Summary

WNS(ns)	TNS(ns)	TNS Failing Endpoints	TNS Total
Endpoints	WHS(ns)	THS(ns)	THS Failing Endpoints
Total Endpoints	WPWS(ns)	TPWS(ns)	TPWS Failing Endpoints
TPWS Total Endpoints			
-----	-----	-----	
-----		-----	-----
-----	-----	-----	-----
-----	-----	-----	-----
3.842	0.000		0
522	0.193	0.000	0
522	4.600	0.000	0
231			

All user specified timing constraints are met.

| Clock Summary

| -----

Clock	Waveform(ns)	Period(ns)	Frequency(MHz)
Clock	{0.000 5.000}	10.000	100.000

| Intra Clock Table

| -----

Clock	WNS(ns)	TNS(ns)	TNS Failing Endpoints	TNS
Total Endpoints	WHS(ns)	THS(ns)	THS Failing Endpoints	
THS Total Endpoints	WPWS(ns)	TPWS(ns)	TPWS Failing	
Endpoints	TPWS Total Endpoints			

Clock	3.842	0.000	0
522	0.193	0.000	0
522	4.600	0.000	0
231			

| Inter Clock Table

| -----

From Clock To Clock WNS(ns) TNS(ns) TNS Failing
Endpoints TNS Total Endpoints WHS(ns) THS(ns) THS
Failing Endpoints THS Total Endpoints

Other Path Groups Table

Path Group From Clock To Clock WNS(ns) TNS(ns)
TNS Failing Endpoints TNS Total Endpoints WHS(ns)
THS(ns) THS Failing Endpoints THS Total Endpoints

User Ignored Path Table

Path Group From Clock To Clock

Unconstrained Path Table

Path Group	From Clock	To Clock
-----	-----	-----
(none)	Clock	
(none)		Clock

Timing Details

From Clock: Clock
To Clock: Clock

Setup : 0 Failing Endpoints, Worst Slack
3.842ns, Total Violation 0.000ns
Hold : 0 Failing Endpoints, Worst Slack
0.193ns, Total Violation 0.000ns
PW : 0 Failing Endpoints, Worst Slack
4.600ns, Total Violation 0.000ns

Max Delay Paths


```

-----
Slack (MET) :          3.842ns  (required time - arrival time)
  Source:          Cycle_Count_reg[0]/C
                  (rising edge-triggered cell FDRE
clocked by Clock  {rise@0.000ns fall@5.000ns period=10.000ns})
  Destination:
SA_Inst/Gen_PE_row[0].Gen_PE_col[0].Gen_Left_Edge.Gen_Top_Left.PE_I
nst/MAC_Unit_Inst/Accumulator_Register_reg/A[5]
                  (rising edge-triggered cell DSP48E1
clocked by Clock  {rise@0.000ns fall@5.000ns period=10.000ns})
  Path Group:          Clock
  Path Type:           Setup (Max at Slow Process Corner)
  Requirement:         10.000ns  (Clock rise@10.000ns - Clock
rise@0.000ns)
  Data Path Delay:      3.009ns  (logic 0.404ns (13.427%)  route
2.605ns (86.573%))
  Logic Levels:        1  (LUT5=1)
  Clock Path Skew:      0.040ns  (DCD - SCD + CPR)
    Destination Clock Delay (DCD):      3.953ns = ( 13.953 - 10.000 )

    Source Clock Delay      (SCD):      4.163ns
    Clock Pessimism Removal (CPR):      0.251ns
  Clock Uncertainty:      0.035ns  ((TSJ^2 + TIJ^2)^1/2 + DJ) / 2 +
PE
    Total System Jitter      (TSJ):      0.071ns
    Total Input Jitter       (TIJ):      0.000ns
    Discrete Jitter          (DJ):      0.000ns
    Phase Error              (PE):      0.000ns

```

Location	Delay type	Incr(ns)
Path(ns)	Netlist Resource(s)	

```

-----
                                (clock Clock rise edge)          0.000
0.000 r

```

D23			0.000
0.000	r	Clock (IN)	
		net (fo=0)	0.000
0.000		Clock	
D23		IBUF (Prop_ibuf_I_O)	0.892
0.892	r	Clock_IBUF_inst/O	
		net (fo=1, routed)	1.764
2.656		Clock_IBUF	
		BUFGCTRL_X0Y0	
		BUFG (Prop_bufg_I_O)	0.120
2.776	r	Clock_IBUF_BUFG_inst/O	
		net (fo=230, routed)	1.387
4.163		Clock_IBUF_BUFG	
		SLICE_X9Y78	
		FDRE	
	r	Cycle_Count_reg[0]/C	

		SLICE_X9Y78	
		FDRE (Prop_fdre_C_Q)	0.246
4.409	r	Cycle_Count_reg[0]/Q	
		net (fo=69, routed)	1.828
6.236			
SA_Inst/Gen_PE_row[0].Gen_PE_col[1].Gen_Interior_Col.Gen_Top_Edge.P			
E_Inst/MAC_Unit_Inst/Q[0]			
		SLICE_X11Y71	
		LUT5 (Prop_lut5_I1_O)	0.158
6.394	r		
SA_Inst/Gen_PE_row[0].Gen_PE_col[1].Gen_Interior_Col.Gen_Top_Edge.P			
E_Inst/MAC_Unit_Inst/Accumulator_Register_reg_i_13/O			
		net (fo=2, routed)	0.777
7.172			
SA_Inst/Gen_PE_row[0].Gen_PE_col[0].Gen_Left_Edge.Gen_Top_Left.PE_I			
nst/MAC_Unit_Inst/A[5]			
		DSP48_X0Y32	
		DSP48E1	
	r		
SA_Inst/Gen_PE_row[0].Gen_PE_col[0].Gen_Left_Edge.Gen_Top_Left.PE_I			
nst/MAC_Unit_Inst/Accumulator_Register_reg/A[5]			

```

-----
                                (clock Clock rise edge)          10.000
10.000 r
    D23                                0.000
10.000 r  Clock (IN)
                                net (fo=0)                    0.000
10.000    Clock
    D23                                IBUF (Prop_ibuf_I_O)      0.815
10.815 r  Clock_IBUF_inst/O
                                net (fo=1, routed)            1.679
12.494    Clock_IBUF
    BUFGCTRL_X0Y0                    BUFG (Prop_bufg_I_O)      0.113
12.607 r  Clock_IBUF_BUFG_inst/O
                                net (fo=230, routed)           1.345
13.953
SA_Inst/Gen_PE_row[0].Gen_PE_col[0].Gen_Left_Edge.Gen_Top_Left.PE_I
nst/MAC_Unit_Inst/CLK
    DSP48_X0Y32                    DSP48E1
r
SA_Inst/Gen_PE_row[0].Gen_PE_col[0].Gen_Left_Edge.Gen_Top_Left.PE_I
nst/MAC_Unit_Inst/Accumulator_Register_reg/CLK
                                clock pessimism                0.251
14.203
                                clock uncertainty              -0.035
14.168
    DSP48_X0Y32                    DSP48E1 (Setup_dsp48e1_CLK_A[5])
                                -3.154
11.014
SA_Inst/Gen_PE_row[0].Gen_PE_col[0].Gen_Left_Edge.Gen_Top_Left.PE_I
nst/MAC_Unit_Inst/Accumulator_Register_reg
-----
                                required time
11.014
                                arrival time
-7.172

```

```

-----
                                slack

3.842

Slack (MET) :                      4.016ns  (required time - arrival time)
  Source:                      FSM_sequential_State_reg[0]/C
                                (rising edge-triggered cell FDRE
clocked by Clock  {rise@0.000ns fall@5.000ns period=10.000ns})
  Destination:
SA_Inst/Gen_PE_row[0].Gen_PE_col[0].Gen_Left_Edge.Gen_Top_Left.PE_I
nst/MAC_Unit_Inst/Accumulator_Register_reg/A[16]
                                (rising edge-triggered cell DSP48E1
clocked by Clock  {rise@0.000ns fall@5.000ns period=10.000ns})
  Path Group:                      Clock
  Path Type:                      Setup (Max at Slow Process Corner)
  Requirement:                      10.000ns  (Clock rise@10.000ns - Clock
rise@0.000ns)
  Data Path Delay:                      2.835ns  (logic 0.361ns (12.732%)  route
2.474ns (87.268%))
  Logic Levels:                      1  (LUT5=1)
  Clock Path Skew:                      0.040ns  (DCD - SCD + CPR)
    Destination Clock Delay (DCD):      3.953ns = ( 13.953 - 10.000 )

    Source Clock Delay      (SCD):      4.163ns
    Clock Pessimism Removal (CPR):      0.251ns
  Clock Uncertainty:      0.035ns  ((TSJ^2 + TIJ^2)^1/2 + DJ) / 2 +
PE

    Total System Jitter      (TSJ):      0.071ns
    Total Input Jitter      (TIJ):      0.000ns
    Discrete Jitter      (DJ):      0.000ns
    Phase Error      (PE):      0.000ns

```

```

Location                      Delay type                      Incr(ns)
Path(ns)      Netlist Resource(s)
-----

```

```

-----
                                (clock Clock rise edge)          0.000
0.000 r
      D23                                                              0.000
0.000 r  Clock (IN)
                                net (fo=0)                      0.000
0.000    Clock
      D23                      IBUF (Prop_ibuf_I_O)              0.892
0.892 r  Clock_IBUF_inst/O
                                net (fo=1, routed)              1.764
2.656    Clock_IBUF
      BUFGCTRL_X0Y0          BUFG (Prop_bufg_I_O)              0.120
2.776 r  Clock_IBUF_BUFG_inst/O
                                net (fo=230, routed)            1.387
4.163    Clock_IBUF_BUFG
      SLICE_X8Y78            FDRE
r  FSM_sequential_State_reg[0]/C
-----

```

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-----
      SLICE_X8Y78            FDRE (Prop_fdre_C_Q)              0.308
4.471 r  FSM_sequential_State_reg[0]/Q
                                net (fo=73, routed)            1.262
5.733
SA_Inst/Gen_PE_row[0].Gen_PE_col[1].Gen_Interior_Col.Gen_Top_Edge.P
E_Inst/MAC_Unit_Inst/State[0]
      SLICE_X10Y73          LUT5 (Prop_lut5_I4_O)              0.053
5.786 r
SA_Inst/Gen_PE_row[0].Gen_PE_col[1].Gen_Interior_Col.Gen_Top_Edge.P
E_Inst/MAC_Unit_Inst/Accumulator_Register_reg_i_11/O
                                net (fo=46, routed)            1.212
6.998
SA_Inst/Gen_PE_row[0].Gen_PE_col[0].Gen_Left_Edge.Gen_Top_Left.PE_I
nst/MAC_Unit_Inst/A[7]
      DSP48_X0Y32            DSP48E1
r
SA_Inst/Gen_PE_row[0].Gen_PE_col[0].Gen_Left_Edge.Gen_Top_Left.PE_I

```

nst/MAC_Unit_Inst/Accumulator_Register_reg/A[16]			

	(clock Clock rise edge)		10.000
10.000 r			
	D23		0.000
10.000 r	Clock (IN)		
	net (fo=0)		0.000
10.000	Clock		
	D23	IBUF (Prop_ibuf_I_O)	0.815
10.815 r	Clock_IBUF_inst/O		
	net (fo=1, routed)		1.679
12.494	Clock_IBUF		
	BUFGCTRL_X0Y0	BUFG (Prop_bufg_I_O)	0.113
12.607 r	Clock_IBUF_BUFG_inst/O		
	net (fo=230, routed)		1.345
13.953			
SA_Inst/Gen_PE_row[0].Gen_PE_col[0].Gen_Left_Edge.Gen_Top_Left.PE_I			
nst/MAC_Unit_Inst/CLK			
	DSP48_X0Y32	DSP48E1	
r			
SA_Inst/Gen_PE_row[0].Gen_PE_col[0].Gen_Left_Edge.Gen_Top_Left.PE_I			
nst/MAC_Unit_Inst/Accumulator_Register_reg/CLK			
	clock pessimism		0.251
14.203			
	clock uncertainty		-0.035
14.168			
	DSP48_X0Y32	DSP48E1 (Setup_dsp48e1_CLK_A[16])	
			-3.154
11.014			
SA_Inst/Gen_PE_row[0].Gen_PE_col[0].Gen_Left_Edge.Gen_Top_Left.PE_I			
nst/MAC_Unit_Inst/Accumulator_Register_reg			

	required time		

```

11.014
                                arrival time
-6.998

-----

                                slack
4.016

Slack (MET) :                  4.016ns  (required time - arrival time)
  Source:                  FSM_sequential_State_reg[0]/C
                              (rising edge-triggered cell FDRE
clocked by Clock  {rise@0.000ns fall@5.000ns period=10.000ns})
  Destination:
SA_Inst/Gen_PE_row[0].Gen_PE_col[0].Gen_Left_Edge.Gen_Top_Left.PE_I
nst/MAC_Unit_Inst/Accumulator_Register_reg/A[17]
                              (rising edge-triggered cell DSP48E1
clocked by Clock  {rise@0.000ns fall@5.000ns period=10.000ns})
  Path Group:                  Clock
  Path Type:                  Setup (Max at Slow Process Corner)
  Requirement:                10.000ns  (Clock rise@10.000ns - Clock
rise@0.000ns)
  Data Path Delay:            2.835ns  (logic 0.361ns (12.732%)  route
2.474ns (87.268%))
  Logic Levels:                1  (LUT5=1)
  Clock Path Skew:            0.040ns  (DCD - SCD + CPR)
    Destination Clock Delay (DCD):      3.953ns = ( 13.953 - 10.000 )

    Source Clock Delay      (SCD):      4.163ns
    Clock Pessimism Removal (CPR):      0.251ns
  Clock Uncertainty:          0.035ns  ((TSJ^2 + TIJ^2)^1/2 + DJ) / 2 +
PE

  Total System Jitter      (TSJ):      0.071ns
  Total Input Jitter       (TIJ):      0.000ns
  Discrete Jitter          (DJ):      0.000ns
  Phase Error              (PE):      0.000ns


Location                    Delay type                    Incr(ns)

```

Path(ns)	Netlist Resource(s)		

	(clock Clock rise edge)	0.000	
0.000 r			
	D23	0.000	
0.000 r	Clock (IN)		
	net (fo=0)	0.000	
0.000	Clock		
	D23	IBUF (Prop_ibuf_I_O)	0.892
0.892 r	Clock_IBUF_inst/O		
	net (fo=1, routed)	1.764	
2.656	Clock_IBUF		
	BUFGCTRL_X0Y0	BUFG (Prop_bufg_I_O)	0.120
2.776 r	Clock_IBUF_BUFG_inst/O		
	net (fo=230, routed)	1.387	
4.163	Clock_IBUF_BUFG		
	SLICE_X8Y78	FDRE	
r	FSM_sequential_State_reg[0]/C		

	SLICE_X8Y78	FDRE (Prop_fdre_C_Q)	0.308
4.471 r	FSM_sequential_State_reg[0]/Q		
	net (fo=73, routed)	1.262	
5.733			
SA_Inst/Gen_PE_row[0].Gen_PE_col[1].Gen_Interior_Col.Gen_Top_Edge.P			
E_Inst/MAC_Unit_Inst/State[0]			
	SLICE_X10Y73	LUT5 (Prop_lut5_I4_O)	0.053
5.786 r			
SA_Inst/Gen_PE_row[0].Gen_PE_col[1].Gen_Interior_Col.Gen_Top_Edge.P			
E_Inst/MAC_Unit_Inst/Accumulator_Register_reg_i_11/O			
	net (fo=46, routed)	1.212	
6.998			
SA_Inst/Gen_PE_row[0].Gen_PE_col[0].Gen_Left_Edge.Gen_Top_Left.PE_I			
nst/MAC_Unit_Inst/A[7]			


```

DSP48_X0Y32          DSP48E1
r
SA_Inst/Gen_PE_row[0].Gen_PE_col[0].Gen_Left_Edge.Gen_Top_Left.PE_I
nst/MAC_Unit_Inst/Accumulator_Register_reg/A[17]
-----
-----
                                (clock Clock rise edge)          10.000
10.000 r
      D23                                0.000
10.000 r  Clock (IN)
                                net (fo=0)                      0.000
10.000      Clock
      D23      IBUF (Prop_ibuf_I_O)          0.815
10.815 r  Clock_IBUF_inst/O
                                net (fo=1, routed)              1.679
12.494      Clock_IBUF
      BUFGCTRL_X0Y0      BUFG (Prop_bufg_I_O)          0.113
12.607 r  Clock_IBUF_BUFG_inst/O
                                net (fo=230, routed)            1.345
13.953
SA_Inst/Gen_PE_row[0].Gen_PE_col[0].Gen_Left_Edge.Gen_Top_Left.PE_I
nst/MAC_Unit_Inst/CLK
      DSP48_X0Y32          DSP48E1
r
SA_Inst/Gen_PE_row[0].Gen_PE_col[0].Gen_Left_Edge.Gen_Top_Left.PE_I
nst/MAC_Unit_Inst/Accumulator_Register_reg/CLK
                                clock pessimism                  0.251
14.203
                                clock uncertainty                 -0.035
14.168
      DSP48_X0Y32          DSP48E1 (Setup_dsp48e1_CLK_A[17])
                                -3.154
11.014
SA_Inst/Gen_PE_row[0].Gen_PE_col[0].Gen_Left_Edge.Gen_Top_Left.PE_I
nst/MAC_Unit_Inst/Accumulator_Register_reg

```

```

-----
                                required time
11.014
                                arrival time
-6.998
-----

                                slack
4.016

Slack (MET) :                      4.016ns  (required time - arrival time)
  Source:                      FSM_sequential_State_reg[0]/C
                                (rising edge-triggered cell FDRE
clocked by Clock  {rise@0.000ns fall@5.000ns period=10.000ns})
  Destination:
SA_Inst/Gen_PE_row[0].Gen_PE_col[0].Gen_Left_Edge.Gen_Top_Left.PE_I
nst/MAC_Unit_Inst/Accumulator_Register_reg/A[18]
                                (rising edge-triggered cell DSP48E1
clocked by Clock  {rise@0.000ns fall@5.000ns period=10.000ns})
  Path Group:                      Clock
  Path Type:                      Setup (Max at Slow Process Corner)
  Requirement:                      10.000ns  (Clock rise@10.000ns - Clock
rise@0.000ns)
  Data Path Delay:                      2.835ns  (logic 0.361ns (12.732%)  route
2.474ns (87.268%))
  Logic Levels:                      1  (LUT5=1)
  Clock Path Skew:                      0.040ns  (DCD - SCD + CPR)
    Destination Clock Delay (DCD):      3.953ns = ( 13.953 - 10.000 )

    Source Clock Delay      (SCD):      4.163ns
    Clock Pessimism Removal (CPR):      0.251ns
  Clock Uncertainty:      0.035ns  ((TSJ^2 + TIJ^2)^1/2 + DJ) / 2 +
PE

  Total System Jitter      (TSJ):      0.071ns
  Total Input Jitter       (TIJ):      0.000ns
  Discrete Jitter          (DJ):      0.000ns

```

Phase Error (PE) : 0.000ns

Location	Delay type	Incr(ns)
Path(ns) Netlist Resource(s)		

	(clock Clock rise edge)	0.000
0.000 r		
D23		0.000
0.000 r	Clock (IN)	
	net (fo=0)	0.000
0.000	Clock	
D23	IBUF (Prop_ibuf_I_O)	0.892
0.892 r	Clock_IBUF_inst/O	
	net (fo=1, routed)	1.764
2.656	Clock_IBUF	
BUFGCTRL_X0Y0	BUFG (Prop_bufg_I_O)	0.120
2.776 r	Clock_IBUF_BUFG_inst/O	
	net (fo=230, routed)	1.387
4.163	Clock_IBUF_BUFG	
SLICE_X8Y78	FDRE	
r	FSM_sequential_State_reg[0]/C	

SLICE_X8Y78	FDRE (Prop_fdre_C_Q)	0.308
4.471 r	FSM_sequential_State_reg[0]/Q	
	net (fo=73, routed)	1.262
5.733		
SA_Inst/Gen_PE_row[0].Gen_PE_col[1].Gen_Interior_Col.Gen_Top_Edge.P		
E_Inst/MAC_Unit_Inst/State[0]		
SLICE_X10Y73	LUT5 (Prop_lut5_I4_O)	0.053
5.786 r		
SA_Inst/Gen_PE_row[0].Gen_PE_col[1].Gen_Interior_Col.Gen_Top_Edge.P		
E_Inst/MAC_Unit_Inst/Accumulator_Register_reg_i_11/O		
	net (fo=46, routed)	1.212

```

6.998
SA_Inst/Gen_PE_row[0].Gen_PE_col[0].Gen_Left_Edge.Gen_Top_Left.PE_I
nst/MAC_Unit_Inst/A[7]
    DSP48_X0Y32          DSP48E1
r
SA_Inst/Gen_PE_row[0].Gen_PE_col[0].Gen_Left_Edge.Gen_Top_Left.PE_I
nst/MAC_Unit_Inst/Accumulator_Register_reg/A[18]
-----
-----
                                (clock Clock rise edge)          10.000
10.000 r
    D23                                0.000
10.000 r  Clock (IN)
                                net (fo=0)          0.000
10.000    Clock
    D23                                IBUF (Prop_ibuf_I_O)      0.815
10.815 r  Clock_IBUF_inst/O
                                net (fo=1, routed)        1.679
12.494    Clock_IBUF
    BUFGCTRL_X0Y0          BUFG (Prop_bufg_I_O)          0.113
12.607 r  Clock_IBUF_BUFG_inst/O
                                net (fo=230, routed)      1.345
13.953
SA_Inst/Gen_PE_row[0].Gen_PE_col[0].Gen_Left_Edge.Gen_Top_Left.PE_I
nst/MAC_Unit_Inst/CLK
    DSP48_X0Y32          DSP48E1
r
SA_Inst/Gen_PE_row[0].Gen_PE_col[0].Gen_Left_Edge.Gen_Top_Left.PE_I
nst/MAC_Unit_Inst/Accumulator_Register_reg/CLK
                                clock pessimism          0.251
14.203
                                clock uncertainty          -0.035
14.168
    DSP48_X0Y32          DSP48E1 (Setup_dsp48e1_CLK_A[18])
                                -3.154

```

```

11.014
SA_Inst/Gen_PE_row[0].Gen_PE_col[0].Gen_Left_Edge.Gen_Top_Left.PE_I
nst/MAC_Unit_Inst/Accumulator_Register_reg
-----
                                required time
11.014
                                arrival time
-6.998
-----
                                slack
4.016

Slack (MET) :                  4.016ns  (required time - arrival time)
  Source:                  FSM_sequential_State_reg[0]/C
                        (rising edge-triggered cell FDRE
clocked by Clock  {rise@0.000ns fall@5.000ns period=10.000ns})
  Destination:
SA_Inst/Gen_PE_row[0].Gen_PE_col[0].Gen_Left_Edge.Gen_Top_Left.PE_I
nst/MAC_Unit_Inst/Accumulator_Register_reg/A[19]
                        (rising edge-triggered cell DSP48E1
clocked by Clock  {rise@0.000ns fall@5.000ns period=10.000ns})
  Path Group:                  Clock
  Path Type:                  Setup (Max at Slow Process Corner)
  Requirement:                10.000ns  (Clock rise@10.000ns - Clock
rise@0.000ns)
  Data Path Delay:            2.835ns  (logic 0.361ns (12.732%)  route
2.474ns (87.268%))
  Logic Levels:                1  (LUT5=1)
  Clock Path Skew:            0.040ns  (DCD - SCD + CPR)
    Destination Clock Delay (DCD):      3.953ns = ( 13.953 - 10.000 )

    Source Clock Delay      (SCD):      4.163ns
    Clock Pessimism Removal (CPR):      0.251ns
  Clock Uncertainty:          0.035ns  ((TSJ^2 + TIJ^2)^1/2 + DJ) / 2 +
PE

```

Total System Jitter	(TSJ):	0.071ns
Total Input Jitter	(TIJ):	0.000ns
Discrete Jitter	(DJ):	0.000ns
Phase Error	(PE):	0.000ns

Location	Delay type	Incr(ns)	
Path(ns)	Netlist Resource(s)		

	(clock Clock rise edge)	0.000	
0.000 r			
D23		0.000	
0.000 r	Clock (IN)		
	net (fo=0)	0.000	
0.000	Clock		
D23	IBUF (Prop_ibuf_I_O)	0.892	
0.892 r	Clock_IBUF_inst/O		
	net (fo=1, routed)	1.764	
2.656	Clock_IBUF		
BUFGCTRL_X0Y0	BUFG (Prop_bufg_I_O)	0.120	
2.776 r	Clock_IBUF_BUFG_inst/O		
	net (fo=230, routed)	1.387	
4.163	Clock_IBUF_BUFG		
SLICE_X8Y78	FDRE		
r	FSM_sequential_State_reg[0]/C		

	SLICE_X8Y78	FDRE (Prop_fdre_C_Q)	0.308
4.471 r	FSM_sequential_State_reg[0]/Q		
	net (fo=73, routed)		1.262
5.733			
SA_Inst/Gen_PE_row[0].Gen_PE_col[1].Gen_Interior_Col.Gen_Top_Edge.P			
E_Inst/MAC_Unit_Inst/State[0]			
	SLICE_X10Y73	LUT5 (Prop_lut5_I4_O)	0.053
5.786 r			

```

SA_Inst/Gen_PE_row[0].Gen_PE_col[1].Gen_Interior_Col.Gen_Top_Edge.P
E_Inst/MAC_Unit_Inst/Accumulator_Register_reg_i_11/O
                                net (fo=46, routed)                                1.212
6.998
SA_Inst/Gen_PE_row[0].Gen_PE_col[0].Gen_Left_Edge.Gen_Top_Left.PE_I
nst/MAC_Unit_Inst/A[7]
    DSP48_X0Y32                DSP48E1
r
SA_Inst/Gen_PE_row[0].Gen_PE_col[0].Gen_Left_Edge.Gen_Top_Left.PE_I
nst/MAC_Unit_Inst/Accumulator_Register_reg/A[19]

-----
-----

                                (clock Clock rise edge)                                10.000
10.000 r
    D23                                                                0.000
10.000 r  Clock (IN)
                                net (fo=0)                                          0.000
10.000    Clock
    D23                        IBUF (Prop_ibuf_I_O)                                0.815
10.815 r  Clock_IBUF_inst/O
                                net (fo=1, routed)                                1.679
12.494    Clock_IBUF
    BUFGCTRL_X0Y0              BUFG (Prop_bufg_I_O)                                0.113
12.607 r  Clock_IBUF_BUFG_inst/O
                                net (fo=230, routed)                                1.345
13.953
SA_Inst/Gen_PE_row[0].Gen_PE_col[0].Gen_Left_Edge.Gen_Top_Left.PE_I
nst/MAC_Unit_Inst/CLK
    DSP48_X0Y32                DSP48E1
r
SA_Inst/Gen_PE_row[0].Gen_PE_col[0].Gen_Left_Edge.Gen_Top_Left.PE_I
nst/MAC_Unit_Inst/Accumulator_Register_reg/CLK
                                clock pessimism                                    0.251
14.203
                                clock uncertainty                                    -0.035

```

```

14.168
    DSP48_X0Y32          DSP48E1 (Setup_dsp48e1_CLK_A[19])
                                -3.154

11.014
SA_Inst/Gen_PE_row[0].Gen_PE_col[0].Gen_Left_Edge.Gen_Top_Left.PE_I
nst/MAC_Unit_Inst/Accumulator_Register_reg

-----
                                required time

11.014
                                arrival time

-6.998

-----
                                slack

4.016

Slack (MET) :                  4.152ns (required time - arrival time)
  Source:                  FSM_sequential_State_reg[0]/C
                        (rising edge-triggered cell FDRE
clocked by Clock {rise@0.000ns fall@5.000ns period=10.000ns})
  Destination:
SA_Inst/Gen_PE_row[0].Gen_PE_col[0].Gen_Left_Edge.Gen_Top_Left.PE_I
nst/MAC_Unit_Inst/Accumulator_Register_reg/A[12]
                        (rising edge-triggered cell DSP48E1
clocked by Clock {rise@0.000ns fall@5.000ns period=10.000ns})
  Path Group:                  Clock
  Path Type:                  Setup (Max at Slow Process Corner)
  Requirement:                10.000ns (Clock rise@10.000ns - Clock
rise@0.000ns)
  Data Path Delay:            2.699ns (logic 0.361ns (13.373%) route
2.338ns (86.627%))
  Logic Levels:                1 (LUT5=1)
  Clock Path Skew:            0.040ns (DCD - SCD + CPR)
    Destination Clock Delay (DCD):      3.953ns = ( 13.953 - 10.000 )

    Source Clock Delay          (SCD):      4.163ns

```



```

Clock Pessimism Removal (CPR):      0.251ns
Clock Uncertainty:      0.035ns  ((TSJ^2 + TIJ^2)^1/2 + DJ) / 2 +
PE

Total System Jitter      (TSJ):      0.071ns
Total Input Jitter      (TIJ):      0.000ns
Discrete Jitter          (DJ):      0.000ns
Phase Error              (PE):      0.000ns


Location                      Delay type                      Incr(ns)
Path(ns)      Netlist Resource(s)

-----
-----
0.000 r                      (clock Clock rise edge)          0.000
    D23                      0.000
0.000 r  Clock (IN)
                      net (fo=0)          0.000
0.000    Clock
    D23                      IBUF (Prop_ibuf_I_O)          0.892
0.892 r  Clock_IBUF_inst/O
                      net (fo=1, routed)          1.764
2.656    Clock_IBUF
    BUFGCTRL_X0Y0          BUFG (Prop_bufg_I_O)          0.120
2.776 r  Clock_IBUF_BUFG_inst/O
                      net (fo=230, routed)          1.387
4.163    Clock_IBUF_BUFG
    SLICE_X8Y78          FDRE
r  FSM_sequential_State_reg[0]/C

-----
-----
    SLICE_X8Y78          FDRE (Prop_fdre_C_Q)          0.308
4.471 r  FSM_sequential_State_reg[0]/Q
                      net (fo=73, routed)          1.262
5.733
SA_Inst/Gen_PE_row[0].Gen_PE_col[1].Gen_Interior_Col.Gen_Top_Edge.P

```

```

E_Inst/MAC_Unit_Inst/State[0]
    SLICE_X10Y73          LUT5 (Prop_lut5_I4_O)          0.053
5.786 r
SA_Inst/Gen_PE_row[0].Gen_PE_col[1].Gen_Interior_Col.Gen_Top_Edge.P
E_Inst/MAC_Unit_Inst/Accumulator_Register_reg_i_11/O
    net (fo=46, routed)          1.076
6.862
SA_Inst/Gen_PE_row[0].Gen_PE_col[0].Gen_Left_Edge.Gen_Top_Left.PE_I
nst/MAC_Unit_Inst/A[7]
    DSP48_X0Y32          DSP48E1
r
SA_Inst/Gen_PE_row[0].Gen_PE_col[0].Gen_Left_Edge.Gen_Top_Left.PE_I
nst/MAC_Unit_Inst/Accumulator_Register_reg/A[12]

-----
-----

    (clock Clock rise edge)          10.000
10.000 r
    D23          0.000
10.000 r  Clock (IN)
    net (fo=0)          0.000
10.000  Clock
    D23          IBUF (Prop_ibuf_I_O)          0.815
10.815 r  Clock_IBUF_inst/O
    net (fo=1, routed)          1.679
12.494  Clock_IBUF
    BUFGCTRL_X0Y0          BUFG (Prop_bufg_I_O)          0.113
12.607 r  Clock_IBUF_BUFG_inst/O
    net (fo=230, routed)          1.345
13.953
SA_Inst/Gen_PE_row[0].Gen_PE_col[0].Gen_Left_Edge.Gen_Top_Left.PE_I
nst/MAC_Unit_Inst/CLK
    DSP48_X0Y32          DSP48E1
r
SA_Inst/Gen_PE_row[0].Gen_PE_col[0].Gen_Left_Edge.Gen_Top_Left.PE_I
nst/MAC_Unit_Inst/Accumulator_Register_reg/CLK

```

```

                                clock pessimism                                0.251
14.203
                                clock uncertainty                                -0.035
14.168
    DSP48_X0Y32                  DSP48E1 (Setup_dsp48e1_CLK_A[12])
                                -3.154
11.014
SA_Inst/Gen_PE_row[0].Gen_PE_col[0].Gen_Left_Edge.Gen_Top_Left.PE_I
nst/MAC_Unit_Inst/Accumulator_Register_reg
-----
                                required time
11.014
                                arrival time
-6.862
-----
                                slack
4.152

Slack (MET) :                  4.152ns  (required time - arrival time)
  Source:                  FSM_sequential_State_reg[0]/C
                            (rising edge-triggered cell FDRE
clocked by Clock  {rise@0.000ns fall@5.000ns period=10.000ns})
  Destination:
SA_Inst/Gen_PE_row[0].Gen_PE_col[0].Gen_Left_Edge.Gen_Top_Left.PE_I
nst/MAC_Unit_Inst/Accumulator_Register_reg/A[13]
                            (rising edge-triggered cell DSP48E1
clocked by Clock  {rise@0.000ns fall@5.000ns period=10.000ns})
  Path Group:                  Clock
  Path Type:                  Setup (Max at Slow Process Corner)
  Requirement:                10.000ns  (Clock rise@10.000ns - Clock
rise@0.000ns)
  Data Path Delay:            2.699ns  (logic 0.361ns (13.373%)  route
2.338ns (86.627%))
  Logic Levels:                1  (LUT5=1)
  Clock Path Skew:            0.040ns  (DCD - SCD + CPR)

```

Destination Clock Delay (DCD): 3.953ns = (13.953 - 10.000)
 Source Clock Delay (SCD): 4.163ns
 Clock Pessimism Removal (CPR): 0.251ns
 Clock Uncertainty: 0.035ns $((TSJ^2 + TIJ^2)^{1/2} + DJ) / 2 + PE$
 Total System Jitter (TSJ): 0.071ns
 Total Input Jitter (TIJ): 0.000ns
 Discrete Jitter (DJ): 0.000ns
 Phase Error (PE): 0.000ns

Location	Delay type	Incr(ns)
Path(ns)	Netlist Resource(s)	

	(clock Clock rise edge)	0.000
0.000 r		
D23		0.000
0.000 r	Clock (IN)	
	net (fo=0)	0.000
0.000	Clock	
D23	IBUF (Prop_ibuf_I_O)	0.892
0.892 r	Clock_IBUF_inst/O	
	net (fo=1, routed)	1.764
2.656	Clock_IBUF	
BUFGCTRL_X0Y0	BUFG (Prop_bufg_I_O)	0.120
2.776 r	Clock_IBUF_BUFG_inst/O	
	net (fo=230, routed)	1.387
4.163	Clock_IBUF_BUFG	
SLICE_X8Y78	FDRE	
r	FSM_sequential_State_reg[0]/C	

SLICE_X8Y78	FDRE (Prop_fdre_C_Q)	0.308
4.471 r	FSM_sequential_State_reg[0]/Q	

```

net (fo=73, routed) 1.262
5.733
SA_Inst/Gen_PE_row[0].Gen_PE_col[1].Gen_Interior_Col.Gen_Top_Edge.P
E_Inst/MAC_Unit_Inst/State[0]
SLICE_X10Y73 LUT5 (Prop_lut5_I4_O) 0.053
5.786 r
SA_Inst/Gen_PE_row[0].Gen_PE_col[1].Gen_Interior_Col.Gen_Top_Edge.P
E_Inst/MAC_Unit_Inst/Accumulator_Register_reg_i_11/O
net (fo=46, routed) 1.076
6.862
SA_Inst/Gen_PE_row[0].Gen_PE_col[0].Gen_Left_Edge.Gen_Top_Left.PE_I
nst/MAC_Unit_Inst/A[7]
DSP48_X0Y32 DSP48E1
r
SA_Inst/Gen_PE_row[0].Gen_PE_col[0].Gen_Left_Edge.Gen_Top_Left.PE_I
nst/MAC_Unit_Inst/Accumulator_Register_reg/A[13]
-----
-----
(clock Clock rise edge) 10.000
10.000 r
D23 0.000
10.000 r Clock (IN)
net (fo=0) 0.000
10.000 Clock
D23 IBUF (Prop_ibuf_I_O) 0.815
10.815 r Clock_IBUF_inst/O
net (fo=1, routed) 1.679
12.494 Clock_IBUF
BUFGCTRL_X0Y0 BUFG (Prop_bufg_I_O) 0.113
12.607 r Clock_IBUF_BUFG_inst/O
net (fo=230, routed) 1.345
13.953
SA_Inst/Gen_PE_row[0].Gen_PE_col[0].Gen_Left_Edge.Gen_Top_Left.PE_I
nst/MAC_Unit_Inst/CLK
DSP48_X0Y32 DSP48E1

```

```

r
SA_Inst/Gen_PE_row[0].Gen_PE_col[0].Gen_Left_Edge.Gen_Top_Left.PE_I
nst/MAC_Unit_Inst/Accumulator_Register_reg/CLK
                                clock pessimism                                0.251
14.203
                                clock uncertainty                                -0.035
14.168
    DSP48_X0Y32                DSP48E1 (Setup_dsp48e1_CLK_A[13])
                                -3.154
11.014
SA_Inst/Gen_PE_row[0].Gen_PE_col[0].Gen_Left_Edge.Gen_Top_Left.PE_I
nst/MAC_Unit_Inst/Accumulator_Register_reg
-----
                                required time
11.014
                                arrival time
-6.862
-----
                                slack
4.152

Slack (MET) :                    4.152ns (required time - arrival time)
Source:                    FSM_sequential_State_reg[0]/C
                        (rising edge-triggered cell FDRE
clocked by Clock {rise@0.000ns fall@5.000ns period=10.000ns})
Destination:
SA_Inst/Gen_PE_row[0].Gen_PE_col[0].Gen_Left_Edge.Gen_Top_Left.PE_I
nst/MAC_Unit_Inst/Accumulator_Register_reg/A[14]
                        (rising edge-triggered cell DSP48E1
clocked by Clock {rise@0.000ns fall@5.000ns period=10.000ns})
Path Group:                    Clock
Path Type:                    Setup (Max at Slow Process Corner)
Requirement:                    10.000ns (Clock rise@10.000ns - Clock
rise@0.000ns)
Data Path Delay:                2.699ns (logic 0.361ns (13.373%) route

```

```
2.338ns (86.627%))
Logic Levels:          1  (LUT5=1)
Clock Path Skew:       0.040ns (DCD - SCD + CPR)
  Destination Clock Delay (DCD):    3.953ns = ( 13.953 - 10.000 )

  Source Clock Delay      (SCD):    4.163ns
  Clock Pessimism Removal (CPR):    0.251ns
Clock Uncertainty:     0.035ns  ((TSJ^2 + TIJ^2)^1/2 + DJ) / 2 +
PE
  Total System Jitter      (TSJ):    0.071ns
  Total Input Jitter       (TIJ):    0.000ns
  Discrete Jitter          (DJ):    0.000ns
  Phase Error              (PE):    0.000ns

  Location                Delay type                Incr(ns)
Path(ns)    Netlist Resource(s)

-----
-----
                                (clock Clock rise edge)                0.000
0.000 r
  D23                                0.000
0.000 r  Clock (IN)
                                net (fo=0)                0.000
0.000    Clock
  D23                                IBUF (Prop_ibuf_I_O)                0.892
0.892 r  Clock_IBUF_inst/O
                                net (fo=1, routed)                1.764
2.656    Clock_IBUF
  BUFGCTRL_X0Y0                    BUFG (Prop_bufg_I_O)                0.120
2.776 r  Clock_IBUF_BUFG_inst/O
                                net (fo=230, routed)                1.387
4.163    Clock_IBUF_BUFG
  SLICE_X8Y78                      FDRE
r  FSM_sequential_State_reg[0]/C
-----
```

```

-----
SLICE_X8Y78          FDRE (Prop_fdre_C_Q)          0.308
4.471 r  FSM_sequential_State_reg[0]/Q
          net (fo=73, routed)          1.262
5.733
SA_Inst/Gen_PE_row[0].Gen_PE_col[1].Gen_Interior_Col.Gen_Top_Edge.P
E_Inst/MAC_Unit_Inst/State[0]
SLICE_X10Y73          LUT5 (Prop_lut5_I4_O)          0.053
5.786 r
SA_Inst/Gen_PE_row[0].Gen_PE_col[1].Gen_Interior_Col.Gen_Top_Edge.P
E_Inst/MAC_Unit_Inst/Accumulator_Register_reg_i_11/O
          net (fo=46, routed)          1.076
6.862
SA_Inst/Gen_PE_row[0].Gen_PE_col[0].Gen_Left_Edge.Gen_Top_Left.PE_I
nst/MAC_Unit_Inst/A[7]
DSP48_X0Y32          DSP48E1
r
SA_Inst/Gen_PE_row[0].Gen_PE_col[0].Gen_Left_Edge.Gen_Top_Left.PE_I
nst/MAC_Unit_Inst/Accumulator_Register_reg/A[14]
-----
-----
          (clock Clock rise edge)          10.000
10.000 r
D23          0.000
10.000 r  Clock (IN)
          net (fo=0)          0.000
10.000      Clock
D23          IBUF (Prop_ibuf_I_O)          0.815
10.815 r  Clock_IBUF_inst/O
          net (fo=1, routed)          1.679
12.494      Clock_IBUF
BUFGCTRL_X0Y0          BUFG (Prop_bufg_I_O)          0.113
12.607 r  Clock_IBUF_BUFG_inst/O
          net (fo=230, routed)          1.345
13.953

```



```

SA_Inst/Gen_PE_row[0].Gen_PE_col[0].Gen_Left_Edge.Gen_Top_Left.PE_I
nst/MAC_Unit_Inst/CLK
    DSP48_X0Y32          DSP48E1
r
SA_Inst/Gen_PE_row[0].Gen_PE_col[0].Gen_Left_Edge.Gen_Top_Left.PE_I
nst/MAC_Unit_Inst/Accumulator_Register_reg/CLK
    clock pessimism          0.251
14.203
    clock uncertainty        -0.035
14.168
    DSP48_X0Y32          DSP48E1 (Setup_dsp48e1_CLK_A[14])
                                -3.154
11.014
SA_Inst/Gen_PE_row[0].Gen_PE_col[0].Gen_Left_Edge.Gen_Top_Left.PE_I
nst/MAC_Unit_Inst/Accumulator_Register_reg
-----
    required time
11.014
    arrival time
-6.862
-----
    slack
4.152

Slack (MET) :          4.152ns (required time - arrival time)
Source:          FSM_sequential_State_reg[0]/C
                  (rising edge-triggered cell FDRE
clocked by Clock {rise@0.000ns fall@5.000ns period=10.000ns})
Destination:
SA_Inst/Gen_PE_row[0].Gen_PE_col[0].Gen_Left_Edge.Gen_Top_Left.PE_I
nst/MAC_Unit_Inst/Accumulator_Register_reg/A[15]
                  (rising edge-triggered cell DSP48E1
clocked by Clock {rise@0.000ns fall@5.000ns period=10.000ns})
Path Group:      Clock
Path Type:       Setup (Max at Slow Process Corner)

```

Requirement: 10.000ns (Clock rise@10.000ns - Clock rise@0.000ns)

Data Path Delay: 2.699ns (logic 0.361ns (13.373%) route 2.338ns (86.627%))

Logic Levels: 1 (LUT5=1)

Clock Path Skew: 0.040ns (DCD - SCD + CPR)

Destination Clock Delay (DCD): 3.953ns = (13.953 - 10.000)

Source Clock Delay (SCD): 4.163ns

Clock Pessimism Removal (CPR): 0.251ns

Clock Uncertainty: 0.035ns ($((TSJ^2 + TIJ^2)^{1/2} + DJ) / 2 + PE$)

Total System Jitter (TSJ): 0.071ns

Total Input Jitter (TIJ): 0.000ns

Discrete Jitter (DJ): 0.000ns

Phase Error (PE): 0.000ns

Location	Delay type	Incr(ns)
Path(ns) Netlist Resource(s)		

	(clock Clock rise edge)	0.000
0.000 r		
D23		0.000
0.000 r Clock (IN)		
	net (fo=0)	0.000
0.000 Clock		
D23	IBUF (Prop_ibuf_I_O)	0.892
0.892 r Clock_IBUF_inst/O		
	net (fo=1, routed)	1.764
2.656 Clock_IBUF		
BUFGCTRL_X0Y0	BUFG (Prop_bufg_I_O)	0.120
2.776 r Clock_IBUF_BUFG_inst/O		
	net (fo=230, routed)	1.387
4.163 Clock_IBUF_BUFG		
SLICE_X8Y78	FDRE	

```

r   FSM_sequential_State_reg[0]/C
-----
-----
SLICE_X8Y78          FDRE  (Prop_fdre_C_Q)          0.308
4.471 r   FSM_sequential_State_reg[0]/Q
          net  (fo=73, routed)          1.262
5.733
SA_Inst/Gen_PE_row[0].Gen_PE_col[1].Gen_Interior_Col.Gen_Top_Edge.P
E_Inst/MAC_Unit_Inst/State[0]
SLICE_X10Y73          LUT5  (Prop_lut5_I4_O)          0.053
5.786 r
SA_Inst/Gen_PE_row[0].Gen_PE_col[1].Gen_Interior_Col.Gen_Top_Edge.P
E_Inst/MAC_Unit_Inst/Accumulator_Register_reg_i_11/O
          net  (fo=46, routed)          1.076
6.862
SA_Inst/Gen_PE_row[0].Gen_PE_col[0].Gen_Left_Edge.Gen_Top_Left.PE_I
nst/MAC_Unit_Inst/A[7]
DSP48_X0Y32          DSP48E1
r
SA_Inst/Gen_PE_row[0].Gen_PE_col[0].Gen_Left_Edge.Gen_Top_Left.PE_I
nst/MAC_Unit_Inst/Accumulator_Register_reg/A[15]
-----
-----
          (clock Clock rise edge)          10.000
10.000 r
D23          0.000
10.000 r   Clock  (IN)
          net  (fo=0)          0.000
10.000      Clock
D23          IBUF  (Prop_ibuf_I_O)          0.815
10.815 r   Clock_IBUF_inst/O
          net  (fo=1, routed)          1.679
12.494      Clock_IBUF
BUFGCTRL_X0Y0          BUFG  (Prop_bufg_I_O)          0.113

```

```

12.607 r   Clock_IBUF_BUF_G_inst/O
              net (fo=230, routed)              1.345
13.953
SA_Inst/Gen_PE_row[0].Gen_PE_col[0].Gen_Left_Edge.Gen_Top_Left.PE_I
nst/MAC_Unit_Inst/CLK
      DSP48_X0Y32      DSP48E1
r
SA_Inst/Gen_PE_row[0].Gen_PE_col[0].Gen_Left_Edge.Gen_Top_Left.PE_I
nst/MAC_Unit_Inst/Accumulator_Register_reg/CLK
              clock pessimism              0.251
14.203
              clock uncertainty              -0.035
14.168
      DSP48_X0Y32      DSP48E1 (Setup_dsp48e1_CLK_A[15])
                              -3.154
11.014
SA_Inst/Gen_PE_row[0].Gen_PE_col[0].Gen_Left_Edge.Gen_Top_Left.PE_I
nst/MAC_Unit_Inst/Accumulator_Register_reg
-----
              required time
11.014
              arrival time
-6.862
-----
              slack
4.152

Slack (MET) :          4.249ns (required time - arrival time)
Source:          FSM_sequential_State_reg[0]/C
                  (rising edge-triggered cell FDRE
clocked by Clock {rise@0.000ns fall@5.000ns period=10.000ns})
Destination:
SA_Inst/Gen_PE_row[0].Gen_PE_col[0].Gen_Left_Edge.Gen_Top_Left.PE_I
nst/MAC_Unit_Inst/Accumulator_Register_reg/A[11]
                  (rising edge-triggered cell DSP48E1

```

```

clocked by Clock {rise@0.000ns fall@5.000ns period=10.000ns})
  Path Group:          Clock
  Path Type:           Setup (Max at Slow Process Corner)
  Requirement:         10.000ns (Clock rise@10.000ns - Clock
rise@0.000ns)
  Data Path Delay:      2.602ns (logic 0.361ns (13.873%) route
2.241ns (86.128%))
  Logic Levels:        1 (LUT5=1)
  Clock Path Skew:      0.040ns (DCD - SCD + CPR)
    Destination Clock Delay (DCD):    3.953ns = ( 13.953 - 10.000 )

    Source Clock Delay (SCD):    4.163ns
    Clock Pessimism Removal (CPR): 0.251ns
  Clock Uncertainty:    0.035ns ((TSJ^2 + TIJ^2)^1/2 + DJ) / 2 +
PE
    Total System Jitter (TSJ):    0.071ns
    Total Input Jitter (TIJ):    0.000ns
    Discrete Jitter (DJ):    0.000ns
    Phase Error (PE):    0.000ns

```

Location	Delay type	Incr(ns)
Path(ns) Netlist Resource(s)		

	(clock Clock rise edge)	0.000
0.000 r		
D23		0.000
0.000 r	Clock (IN)	
	net (fo=0)	0.000
0.000	Clock	
D23	IBUF (Prop_ibuf_I_O)	0.892
0.892 r	Clock_IBUF_inst/O	
	net (fo=1, routed)	1.764
2.656	Clock_IBUF	
BUFGCTRL_X0Y0	BUFG (Prop_bufg_I_O)	0.120
2.776 r	Clock_IBUF_BUFG_inst/O	

```

                                net (fo=230, routed)                1.387
4.163      Clock_IBUF_BUFG
          SLICE_X8Y78          FDRE
r  FSM_sequential_State_reg[0]/C
-----
-----
          SLICE_X8Y78          FDRE (Prop_fdre_C_Q)                0.308
4.471 r  FSM_sequential_State_reg[0]/Q
                                net (fo=73, routed)                1.262
5.733
SA_Inst/Gen_PE_row[0].Gen_PE_col[1].Gen_Interior_Col.Gen_Top_Edge.P
E_Inst/MAC_Unit_Inst/State[0]
          SLICE_X10Y73          LUT5 (Prop_lut5_I4_O)                0.053
5.786 r
SA_Inst/Gen_PE_row[0].Gen_PE_col[1].Gen_Interior_Col.Gen_Top_Edge.P
E_Inst/MAC_Unit_Inst/Accumulator_Register_reg_i_11/O
                                net (fo=46, routed)                0.979
6.765
SA_Inst/Gen_PE_row[0].Gen_PE_col[0].Gen_Left_Edge.Gen_Top_Left.PE_I
nst/MAC_Unit_Inst/A[7]
          DSP48_X0Y32          DSP48E1
r
SA_Inst/Gen_PE_row[0].Gen_PE_col[0].Gen_Left_Edge.Gen_Top_Left.PE_I
nst/MAC_Unit_Inst/Accumulator_Register_reg/A[11]
-----
-----
                                (clock Clock rise edge)            10.000
10.000 r
          D23                                0.000
10.000 r  Clock (IN)
                                net (fo=0)                        0.000
10.000      Clock
          D23          IBUF (Prop_ibuf_I_O)                0.815
10.815 r  Clock_IBUF_inst/O

```

	net (fo=1, routed)	1.679
12.494	Clock_IBUF	
	BUFGCTRL_X0Y0	BUFG (Prop_bufg_I_O)
		0.113
12.607	r Clock_IBUF_BUFG_inst/O	
	net (fo=230, routed)	1.345
13.953		
	SA_Inst/Gen_PE_row[0].Gen_PE_col[0].Gen_Left_Edge.Gen_Top_Left.PE_I	
	nst/MAC_Unit_Inst/CLK	
	DSP48_X0Y32	DSP48E1
	r	
	SA_Inst/Gen_PE_row[0].Gen_PE_col[0].Gen_Left_Edge.Gen_Top_Left.PE_I	
	nst/MAC_Unit_Inst/Accumulator_Register_reg/CLK	
	clock pessimism	0.251
14.203		
	clock uncertainty	-0.035
14.168		
	DSP48_X0Y32	DSP48E1 (Setup_dsp48e1_CLK_A[11])
		-3.154
11.014		
	SA_Inst/Gen_PE_row[0].Gen_PE_col[0].Gen_Left_Edge.Gen_Top_Left.PE_I	
	nst/MAC_Unit_Inst/Accumulator_Register_reg	

	required time	
11.014		
	arrival time	
-6.765		

	slack	
4.249		

Min Delay Paths

```

-----
Slack (MET) :          0.193ns  (arrival time - required time)
Source:          Cycle_Count_reg[0]/C
                  (rising edge-triggered cell FDRE
clocked by Clock  {rise@0.000ns fall@5.000ns period=10.000ns})
Destination:     Cycle_Count_reg[13]/D
                  (rising edge-triggered cell FDRE
clocked by Clock  {rise@0.000ns fall@5.000ns period=10.000ns})
Path Group:      Clock
Path Type:       Hold (Min at Fast Process Corner)
Requirement:     0.000ns  (Clock rise@0.000ns - Clock
rise@0.000ns)
Data Path Delay:  0.253ns  (logic 0.157ns (62.067%)  route
0.096ns (37.933%))
Logic Levels:    1  (LUT6=1)
Clock Path Skew:  0.000ns  (DCD - SCD - CPR)
  Destination Clock Delay (DCD):    1.912ns
  Source Clock Delay      (SCD):    1.398ns
  Clock Pessimism Removal (CPR):    0.514ns

```

Location	Delay type	Incr(ns)
Path(ns)	Netlist Resource(s)	

	(clock Clock rise edge)	0.000
0.000 r		
D23		0.000
0.000 r	Clock (IN)	
	net (fo=0)	0.000
0.000	Clock	
D23	IBUF (Prop_ibuf_I_O)	0.155
0.155 r	Clock_IBUF_inst/O	
	net (fo=1, routed)	0.676
0.831	Clock_IBUF	
BUFGCTRL_X0Y0	BUFG (Prop_bufg_I_O)	0.026

0.857	r	Clock_IBUF_BUFG_inst/O		
		net (fo=230, routed)		0.541
1.398		Clock_IBUF_BUFG		
		SLICE_X9Y78	FDRE	
	r	Cycle_Count_reg[0]/C		

		SLICE_X9Y78	FDRE (Prop_fdre_C_Q)	0.091
1.489	f	Cycle_Count_reg[0]/Q		
		net (fo=69, routed)		0.096
1.585		Cycle_Count_reg_n_0_[0]		
		SLICE_X9Y78	LUT6 (Prop_lut6_I1_O)	0.066
1.651	r	Cycle_Count[13]_i_1/O		
		net (fo=1, routed)		0.000
1.651		Cycle_Count[13]_i_1_n_0		
		SLICE_X9Y78	FDRE	
	r	Cycle_Count_reg[13]/D		

			(clock Clock rise edge)	0.000
0.000	r			
		D23		0.000
0.000	r	Clock (IN)		
		net (fo=0)		0.000
0.000		Clock		
		D23	IBUF (Prop_ibuf_I_O)	0.404
0.404	r	Clock_IBUF_inst/O		
		net (fo=1, routed)		0.739
1.143		Clock_IBUF		
		BUFGCTRL_X0Y0	BUFG (Prop_bufg_I_O)	0.030
1.173	r	Clock_IBUF_BUFG_inst/O		
		net (fo=230, routed)		0.739
1.912		Clock_IBUF_BUFG		
		SLICE_X9Y78	FDRE	

```

r   Cycle_Count_reg[13]/C
                                clock pessimism                                -0.514
1.398
    SLICE_X9Y78                FDRE (Hold_fdre_C_D)                0.060
1.458    Cycle_Count_reg[13]

-----

                                required time
-1.458
                                arrival time
1.651

-----

                                slack
0.193

Slack (MET) :                  0.221ns  (arrival time - required time)
Source:                  FSM_sequential_State_reg[1]/C
                        (rising edge-triggered cell FDRE
clocked by Clock {rise@0.000ns fall@5.000ns period=10.000ns})
Destination:            Cycle_Count_reg[12]/D
                        (rising edge-triggered cell FDRE
clocked by Clock {rise@0.000ns fall@5.000ns period=10.000ns})
Path Group:              Clock
Path Type:               Hold (Min at Fast Process Corner)
Requirement:             0.000ns  (Clock rise@0.000ns - Clock
rise@0.000ns)
Data Path Delay:         0.310ns  (logic 0.171ns (55.081%)  route
0.139ns (44.919%))
Logic Levels:            1  (LUT6=1)
Clock Path Skew:         0.029ns  (DCD - SCD - CPR)
  Destination Clock Delay (DCD):      1.911ns
  Source Clock Delay      (SCD):      1.398ns
  Clock Pessimism Removal (CPR):      0.484ns

Location                  Delay type                  Incr(ns)
Path(ns)    Netlist Resource(s)

```

```

-----
                                (clock Clock rise edge)          0.000
0.000 r
    D23                                0.000
0.000 r  Clock (IN)
                                net (fo=0)                    0.000
0.000    Clock
    D23                                IBUF (Prop_ibuf_I_O)      0.155
0.155 r  Clock_IBUF_inst/O
                                net (fo=1, routed)            0.676
0.831    Clock_IBUF
    BUFGCTRL_X0Y0                    BUFG (Prop_bufg_I_O)      0.026
0.857 r  Clock_IBUF_BUFG_inst/O
                                net (fo=230, routed)           0.541
1.398    Clock_IBUF_BUFG
    SLICE_X8Y78                      FDRE
r  FSM_sequential_State_reg[1]/C

```

```

-----
    SLICE_X8Y78                      FDRE (Prop_fdre_C_Q)      0.107
1.505 r  FSM_sequential_State_reg[1]/Q
                                net (fo=72, routed)           0.139
1.645    State[1]
    SLICE_X11Y77                      LUT6 (Prop_lut6_I0_O)    0.064
1.709 r  Cycle_Count[12]_i_1/O
                                net (fo=1, routed)             0.000
1.709    Cycle_Count[12]_i_1_n_0
    SLICE_X11Y77                      FDRE
r  Cycle_Count_reg[12]/D

```

```

-----
                                (clock Clock rise edge)          0.000

```

0.000	r				
		D23			0.000
0.000	r	Clock (IN)			
			net (fo=0)		0.000
0.000		Clock			
		D23	IBUF (Prop_ibuf_I_O)		0.404
0.404	r	Clock_IBUF_inst/O			
			net (fo=1, routed)		0.739
1.143		Clock_IBUF			
		BUFGCTRL_X0Y0	BUFG (Prop_bufg_I_O)		0.030
1.173	r	Clock_IBUF_BUFG_inst/O			
			net (fo=230, routed)		0.738
1.911		Clock_IBUF_BUFG			
		SLICE_X11Y77	FDRE		
r		Cycle_Count_reg[12]/C			
			clock pessimism		-0.484
1.427					
		SLICE_X11Y77	FDRE (Hold_fdre_C_D)		0.060
1.487		Cycle_Count_reg[12]			

			required time		
-1.487					
			arrival time		
1.709					

			slack		
0.221					
Slack (MET) :			0.235ns	(arrival time - required time)	
Source:			Cycle_Count_reg[1]/C		
			(rising edge-triggered cell FDRE		
clocked by Clock		{rise@0.000ns fall@5.000ns period=10.000ns}}			
Destination:			Cycle_Count_reg[3]/D		
			(rising edge-triggered cell FDRE		
clocked by Clock		{rise@0.000ns fall@5.000ns period=10.000ns}}			

Path Group: Clock
 Path Type: Hold (Min at Fast Process Corner)
 Requirement: 0.000ns (Clock rise@0.000ns - Clock
 rise@0.000ns)
 Data Path Delay: 0.296ns (logic 0.128ns (43.305%) route
 0.168ns (56.695%))
 Logic Levels: 1 (LUT6=1)
 Clock Path Skew: 0.000ns (DCD - SCD - CPR)
 Destination Clock Delay (DCD): 1.908ns
 Source Clock Delay (SCD): 1.395ns
 Clock Pessimism Removal (CPR): 0.513ns

Location	Delay type	Incr(ns)
Path(ns)	Netlist Resource(s)	

	(clock Clock rise edge)	0.000
0.000 r		
D23		0.000
0.000 r	Clock (IN)	
	net (fo=0)	0.000
0.000	Clock	
D23	IBUF (Prop_ibuf_I_O)	0.155
0.155 r	Clock_IBUF_inst/O	
	net (fo=1, routed)	0.676
0.831	Clock_IBUF	
BUFGCTRL_X0Y0	BUFG (Prop_bufg_I_O)	0.026
0.857 r	Clock_IBUF_BUFG_inst/O	
	net (fo=230, routed)	0.538
1.395	Clock_IBUF_BUFG	
SLICE_X11Y75	FDRE	
r	Cycle_Count_reg[1]/C	

SLICE_X11Y75	FDRE (Prop_fdre_C_Q)	0.100
--------------	----------------------	-------

1.495	r	Cycle_Count_reg[1]/Q		
		net (fo=36, routed)		0.168
1.663		Cycle_Count_reg_n_0_[1]		
		SLICE_X11Y75	LUT6 (Prop_lut6_I2_O)	0.028
1.691	r	Cycle_Count[3]_i_1/O		
		net (fo=1, routed)		0.000
1.691		Cycle_Count[3]_i_1_n_0		
		SLICE_X11Y75	FDRE	
	r	Cycle_Count_reg[3]/D		

			(clock Clock rise edge)	0.000
0.000	r	D23		
				0.000
0.000	r	Clock (IN)		
		net (fo=0)		0.000
0.000		Clock		
		D23	IBUF (Prop_ibuf_I_O)	0.404
0.404	r	Clock_IBUF_inst/O		
		net (fo=1, routed)		0.739
1.143		Clock_IBUF		
		BUFGCTRL_X0Y0	BUFG (Prop_bufg_I_O)	0.030
1.173	r	Clock_IBUF_BUFG_inst/O		
		net (fo=230, routed)		0.735
1.908		Clock_IBUF_BUFG		
		SLICE_X11Y75	FDRE	
	r	Cycle_Count_reg[3]/C		
		clock pessimism		-0.513
1.395				
		SLICE_X11Y75	FDRE (Hold_fdre_C_D)	0.061
1.456		Cycle_Count_reg[3]		

		required time		
-1.456				

```

                                arrival time
1.691

-----

                                slack
0.235

Slack (MET) :                  0.237ns  (arrival time - required time)
Source:                  Cycle_Count_reg[1]/C
                        (rising edge-triggered cell FDRE
clocked by Clock {rise@0.000ns fall@5.000ns period=10.000ns})
Destination:            Cycle_Count_reg[4]/D
                        (rising edge-triggered cell FDRE
clocked by Clock {rise@0.000ns fall@5.000ns period=10.000ns})
Path Group:              Clock
Path Type:               Hold (Min at Fast Process Corner)
Requirement:             0.000ns  (Clock rise@0.000ns - Clock
rise@0.000ns)
Data Path Delay:         0.298ns  (logic 0.128ns (43.014%)  route
0.170ns (56.986%))
Logic Levels:            1  (LUT6=1)
Clock Path Skew:         0.000ns  (DCD - SCD - CPR)
  Destination Clock Delay (DCD):      1.908ns
  Source Clock Delay      (SCD):      1.395ns
  Clock Pessimism Removal (CPR):      0.513ns


Location                  Delay type                  Incr(ns)
Path(ns)      Netlist Resource(s)

-----
                                (clock Clock rise edge)      0.000
0.000 r
  D23
0.000 r  Clock (IN)
                                net (fo=0)      0.000
0.000      Clock

```

	D23	IBUF (Prop_ibuf_I_O)	0.155
0.155	r	Clock_IBUF_inst/O	
		net (fo=1, routed)	0.676
0.831		Clock_IBUF	
	BUFGCTRL_X0Y0	BUFG (Prop_bufg_I_O)	0.026
0.857	r	Clock_IBUF_BUFG_inst/O	
		net (fo=230, routed)	0.538
1.395		Clock_IBUF_BUFG	
	SLICE_X11Y75	FDRE	
	r	Cycle_Count_reg[1]/C	

	SLICE_X11Y75	FDRE (Prop_fdre_C_Q)	0.100
1.495	r	Cycle_Count_reg[1]/Q	
		net (fo=36, routed)	0.170
1.665		Cycle_Count_reg_n_0_[1]	
	SLICE_X11Y75	LUT6 (Prop_lut6_I2_O)	0.028
1.693	r	Cycle_Count[4]_i_1/O	
		net (fo=1, routed)	0.000
1.693		Cycle_Count[4]_i_1_n_0	
	SLICE_X11Y75	FDRE	
	r	Cycle_Count_reg[4]/D	

		(clock Clock rise edge)	0.000
0.000	r		
	D23		0.000
0.000	r	Clock (IN)	
		net (fo=0)	0.000
0.000		Clock	
	D23	IBUF (Prop_ibuf_I_O)	0.404
0.404	r	Clock_IBUF_inst/O	
		net (fo=1, routed)	0.739
1.143		Clock_IBUF	

	BUFGCTRL_X0Y0	BUFG (Prop_bufg_I_O)	0.030
1.173	r	Clock_IBUF_BUFG_inst/O	
		net (fo=230, routed)	0.735
1.908		Clock_IBUF_BUFG	
		SLICE_X11Y75	FDRE
	r	Cycle_Count_reg[4]/C	
		clock pessimism	-0.513
1.395			
		SLICE_X11Y75	FDRE (Hold_fdre_C_D)
			0.061
1.456		Cycle_Count_reg[4]	

		required time	
-1.456			
		arrival time	
1.693			

		slack	
0.237			

Slack (MET) : 0.240ns (arrival time - required time)

Source: Cycle_Count_reg[1]/C
(rising edge-triggered cell FDRE

clocked by Clock {rise@0.000ns fall@5.000ns period=10.000ns})

Destination: Cycle_Count_reg[1]/D
(rising edge-triggered cell FDRE

clocked by Clock {rise@0.000ns fall@5.000ns period=10.000ns})

Path Group: Clock

Path Type: Hold (Min at Fast Process Corner)

Requirement: 0.000ns (Clock rise@0.000ns - Clock
rise@0.000ns)

Data Path Delay: 0.300ns (logic 0.128ns (42.611%) route
0.172ns (57.389%))

Logic Levels: 1 (LUT6=1)

Clock Path Skew: 0.000ns (DCD - SCD - CPR)

Destination Clock Delay (DCD): 1.908ns

Source Clock Delay (SCD): 1.395ns

Clock Pessimism Removal (CPR): 0.513ns

Location	Delay type	Incr(ns)
Path(ns)	Netlist Resource(s)	

	(clock Clock rise edge)	0.000
0.000 r		
D23		0.000
0.000 r	Clock (IN)	
	net (fo=0)	0.000
0.000	Clock	
D23	IBUF (Prop_ibuf_I_O)	0.155
0.155 r	Clock_IBUF_inst/O	
	net (fo=1, routed)	0.676
0.831	Clock_IBUF	
BUFGCTRL_X0Y0	BUFG (Prop_bufg_I_O)	0.026
0.857 r	Clock_IBUF_BUFG_inst/O	
	net (fo=230, routed)	0.538
1.395	Clock_IBUF_BUFG	
SLICE_X11Y75	FDRE	
r	Cycle_Count_reg[1]/C	

SLICE_X11Y75	FDRE (Prop_fdre_C_Q)	0.100
1.495 r	Cycle_Count_reg[1]/Q	
	net (fo=36, routed)	0.172
1.668	Cycle_Count_reg_n_0[1]	
SLICE_X11Y75	LUT6 (Prop_lut6_I2_O)	0.028
1.696 r	Cycle_Count[1]_i_1/O	
	net (fo=1, routed)	0.000
1.696	Cycle_Count[1]_i_1_n_0	
SLICE_X11Y75	FDRE	
r	Cycle_Count_reg[1]/D	

```

-----
                                (clock Clock rise edge)          0.000
0.000 r
      D23                                                         0.000
0.000 r  Clock (IN)
                                net (fo=0)                      0.000
0.000      Clock
      D23      IBUF (Prop_ibuf_I_O)          0.404
0.404 r  Clock_IBUF_inst/O
                                net (fo=1, routed)              0.739
1.143      Clock_IBUF
      BUFGCTRL_X0Y0      BUFG (Prop_bufg_I_O)          0.030
1.173 r  Clock_IBUF_BUFG_inst/O
                                net (fo=230, routed)            0.735
1.908      Clock_IBUF_BUFG
      SLICE_X11Y75      FDRE
r  Cycle_Count_reg[1]/C
                                clock pessimism                -0.513
1.395
      SLICE_X11Y75      FDRE (Hold_fdre_C_D)          0.060
1.455      Cycle_Count_reg[1]

```

```

-----
                                required time
-1.455
                                arrival time
1.696

```

```

-----
                                slack
0.240

```

```

Slack (MET) :          0.241ns  (arrival time - required time)
Source:          B_Reg_reg[0][1][6]/C

```

```

(rising edge-triggered cell FDRE
clocked by Clock {rise@0.000ns fall@5.000ns period=10.000ns})
  Destination:
SA_Inst/Gen_PE_row[0].Gen_PE_col[1].Gen_Interior_Col.Gen_Top_Edge.P
E_Inst/MAC_Unit_Inst/Accumulator_Register_reg/B[6]
(rising edge-triggered cell DSP48E1
clocked by Clock {rise@0.000ns fall@5.000ns period=10.000ns})
  Path Group:          Clock
  Path Type:           Hold (Min at Fast Process Corner)
  Requirement:         0.000ns (Clock rise@0.000ns - Clock
rise@0.000ns)
  Data Path Delay:      0.410ns (logic 0.146ns (35.590%) route
0.264ns (64.410%))
  Logic Levels:        1 (LUT5=1)
  Clock Path Skew:      0.099ns (DCD - SCD - CPR)
    Destination Clock Delay (DCD): 1.981ns
    Source Clock Delay (SCD): 1.398ns
    Clock Pessimism Removal (CPR): 0.484ns

```

Location	Delay type	Incr(ns)
Path(ns) Netlist Resource(s)		

	(clock Clock rise edge)	0.000
0.000 r		
D23		0.000
0.000 r Clock (IN)		
	net (fo=0)	0.000
0.000 Clock		
D23	IBUF (Prop_ibuf_I_O)	0.155
0.155 r Clock_IBUF_inst/O		
	net (fo=1, routed)	0.676
0.831 Clock_IBUF		
BUFGCTRL_X0Y0	BUFG (Prop_bufg_I_O)	0.026
0.857 r Clock_IBUF_BUFG_inst/O		
	net (fo=230, routed)	0.541

```

1.398      Clock_IBUF_BUFG
          SLICE_X12Y79          FDRE
r   B_Reg_reg[0][1][6]/C

-----

          SLICE_X12Y79          FDRE (Prop_fdre_C_Q)          0.118
1.516 r   B_Reg_reg[0][1][6]/Q
          net (fo=1, routed)          0.052

1.569
SA_Inst/Gen_PE_row[1].Gen_PE_col[1].Gen_Interior_Col.Gen_Interior.P
E_Inst/MAC_Unit_Inst/Accumulator_Register_reg_2[6]
          SLICE_X13Y79          LUT5 (Prop_lut5_I3_O)          0.028
1.597 r
SA_Inst/Gen_PE_row[1].Gen_PE_col[1].Gen_Interior_Col.Gen_Interior.P
E_Inst/MAC_Unit_Inst/Accumulator_Register_reg_i_2__0/O
          net (fo=2, routed)          0.212

1.809
SA_Inst/Gen_PE_row[0].Gen_PE_col[1].Gen_Interior_Col.Gen_Top_Edge.P
E_Inst/MAC_Unit_Inst/B[6]
          DSP48_X0Y33          DSP48E1
r
SA_Inst/Gen_PE_row[0].Gen_PE_col[1].Gen_Interior_Col.Gen_Top_Edge.P
E_Inst/MAC_Unit_Inst/Accumulator_Register_reg/B[6]

-----

          (clock Clock rise edge)          0.000
0.000 r
          D23          0.000
0.000 r   Clock (IN)
          net (fo=0)          0.000

0.000      Clock
          D23          IBUF (Prop_ibuf_I_O)          0.404
0.404 r   Clock_IBUF_inst/O
          net (fo=1, routed)          0.739

```

```

1.143      Clock_IBUF
          BUFGCTRL_X0Y0          BUFG (Prop_bufg_I_O)          0.030
1.173 r    Clock_IBUF_BUFG_inst/O
          net (fo=230, routed)          0.808
1.981
SA_Inst/Gen_PE_row[0].Gen_PE_col[1].Gen_Interior_Col.Gen_Top_Edge.P
E_Inst/MAC_Unit_Inst/CLK
          DSP48_X0Y33          DSP48E1
r
SA_Inst/Gen_PE_row[0].Gen_PE_col[1].Gen_Interior_Col.Gen_Top_Edge.P
E_Inst/MAC_Unit_Inst/Accumulator_Register_reg/CLK
          clock pessimism          -0.484
1.498
          DSP48_X0Y33          DSP48E1 (Hold_dsp48e1_CLK_B[6])
          0.070
1.568
SA_Inst/Gen_PE_row[0].Gen_PE_col[1].Gen_Interior_Col.Gen_Top_Edge.P
E_Inst/MAC_Unit_Inst/Accumulator_Register_reg
-----
          required time
-1.568
          arrival time
1.809
-----
          slack
0.241

Slack (MET) :          0.246ns (arrival time - required time)
Source:          Cycle_Count_reg[0]/C
                  (rising edge-triggered cell FDRE
clocked by Clock {rise@0.000ns fall@5.000ns period=10.000ns})
Destination:     Cycle_Count_reg[11]/D
                  (rising edge-triggered cell FDRE
clocked by Clock {rise@0.000ns fall@5.000ns period=10.000ns})
Path Group:      Clock

```

Path Type: Hold (Min at Fast Process Corner)
 Requirement: 0.000ns (Clock rise@0.000ns - Clock
 rise@0.000ns)
 Data Path Delay: 0.317ns (logic 0.157ns (49.555%) route
 0.160ns (50.445%))
 Logic Levels: 1 (LUT6=1)
 Clock Path Skew: 0.011ns (DCD - SCD - CPR)
 Destination Clock Delay (DCD): 1.911ns
 Source Clock Delay (SCD): 1.398ns
 Clock Pessimism Removal (CPR): 0.502ns

Location	Delay type	Incr(ns)
Path(ns) Netlist Resource(s)		

	(clock Clock rise edge)	0.000
0.000 r		
D23		0.000
0.000 r Clock (IN)		
	net (fo=0)	0.000
0.000 Clock		
D23	IBUF (Prop_ibuf_I_O)	0.155
0.155 r Clock_IBUF_inst/O		
	net (fo=1, routed)	0.676
0.831 Clock_IBUF		
BUFGCTRL_X0Y0	BUFG (Prop_bufg_I_O)	0.026
0.857 r Clock_IBUF_BUFG_inst/O		
	net (fo=230, routed)	0.541
1.398 Clock_IBUF_BUFG		
SLICE_X9Y78	FDRE	
r Cycle_Count_reg[0]/C		

SLICE_X9Y78	FDRE (Prop_fdre_C_Q)	0.091
1.489 f Cycle_Count_reg[0]/Q		

	net (fo=69, routed)	0.160
1.649	Cycle_Count_reg_n_0_[0]	
	SLICE_X9Y77 LUT6 (Prop_lut6_I1_O)	0.066
1.715	r Cycle_Count[11]_i_1/O	
	net (fo=1, routed)	0.000
1.715	Cycle_Count[11]_i_1_n_0	
	SLICE_X9Y77 FDRE	
	r Cycle_Count_reg[11]/D	

	(clock Clock rise edge)	0.000
0.000	r	
	D23	0.000
0.000	r Clock (IN)	
	net (fo=0)	0.000
0.000	Clock	
	D23 IBUF (Prop_ibuf_I_O)	0.404
0.404	r Clock_IBUF_inst/O	
	net (fo=1, routed)	0.739
1.143	Clock_IBUF	
	BUFGCTRL_X0Y0 BUFG (Prop_bufg_I_O)	0.030
1.173	r Clock_IBUF_BUFG_inst/O	
	net (fo=230, routed)	0.738
1.911	Clock_IBUF_BUFG	
	SLICE_X9Y77 FDRE	
	r Cycle_Count_reg[11]/C	
	clock pessimism	-0.502
1.409		
	SLICE_X9Y77 FDRE (Hold_fdre_C_D)	0.060
1.469	Cycle_Count_reg[11]	

	required time	
-1.469		
	arrival time	

1.715

slack
0.246

Slack (MET) : 0.246ns (arrival time - required time)

Source:

SA_Inst/Gen_PE_row[1].Gen_PE_col[0].Gen_Left_Edge.Gen_Left_Interior
.PE_Inst/MAC_Unit_Inst/Accumulator_Register_reg/CLK

(rising edge-triggered cell DSP48E1

clocked by Clock {rise@0.000ns fall@5.000ns period=10.000ns})

Destination: Output_Reg_reg[1][0][7]/D

(rising edge-triggered cell FDRE

clocked by Clock {rise@0.000ns fall@5.000ns period=10.000ns})

Path Group: Clock

Path Type: Hold (Min at Fast Process Corner)

Requirement: 0.000ns (Clock rise@0.000ns - Clock
rise@0.000ns)

Data Path Delay: 0.270ns (logic 0.107ns (39.634%) route
0.163ns (60.366%))

Logic Levels: 0

Clock Path Skew: -0.035ns (DCD - SCD - CPR)

Destination Clock Delay (DCD): 1.909ns

Source Clock Delay (SCD): 1.461ns

Clock Pessimism Removal (CPR): 0.484ns

Location	Delay type	Incr(ns)
Path(ns)	Netlist Resource(s)	

(clock Clock rise edge) 0.000
0.000 r
D23 0.000
0.000 r Clock (IN)
net (fo=0) 0.000

```

0.000      Clock
      D23      IBUF (Prop_ibuf_I_O)      0.155
0.155 r  Clock_IBUF_inst/O
      net (fo=1, routed)      0.676
0.831      Clock_IBUF
      BUFGCTRL_X0Y0      BUFG (Prop_bufg_I_O)      0.026
0.857 r  Clock_IBUF_BUFG_inst/O
      net (fo=230, routed)      0.603
1.461
SA_Inst/Gen_PE_row[1].Gen_PE_col[0].Gen_Left_Edge.Gen_Left_Interior
.PE_Inst/MAC_Unit_Inst/CLK
      DSP48_X0Y30      DSP48E1
r
SA_Inst/Gen_PE_row[1].Gen_PE_col[0].Gen_Left_Edge.Gen_Left_Interior
.PE_Inst/MAC_Unit_Inst/Accumulator_Register_reg/CLK
-----
-----
      DSP48_X0Y30      DSP48E1 (Prop_dsp48e1_CLK_P[7])
      0.107
1.568 r
SA_Inst/Gen_PE_row[1].Gen_PE_col[0].Gen_Left_Edge.Gen_Left_Interior
.PE_Inst/MAC_Unit_Inst/Accumulator_Register_reg/P[7]
      net (fo=1, routed)      0.163
1.731      SA_Inst_n_88
      SLICE_X12Y76      FDRE
r  Output_Reg_reg[1][0][7]/D
-----
-----
      (clock Clock rise edge)      0.000
0.000 r
      D23      0.000
0.000 r  Clock (IN)
      net (fo=0)      0.000
0.000      Clock

```

D23	IBUF (Prop_ibuf_I_O)	0.404
0.404 r	Clock_IBUF_inst/O	
	net (fo=1, routed)	0.739
1.143	Clock_IBUF	
	BUFGCTRL_X0Y0	0.030
1.173 r	Clock_IBUF_BUFG_inst/O	
	net (fo=230, routed)	0.736
1.909	Clock_IBUF_BUFG	
	SLICE_X12Y76	
	FDRE	
r	Output_Reg_reg[1][0][7]/C	
	clock pessimism	-0.484
1.425		
	SLICE_X12Y76	
	FDRE (Hold_fdre_C_D)	0.059
1.484	Output_Reg_reg[1][0][7]	

	required time	
-1.484		
	arrival time	
1.731		

	slack	
0.246		
Slack (MET) : 0.254ns (arrival time - required time)		
Source:	Cycle_Count_reg[0]/C	
	(rising edge-triggered cell FDRE	
clocked by Clock	{rise@0.000ns fall@5.000ns period=10.000ns})	
Destination:	Cycle_Count_reg[18]/D	
	(rising edge-triggered cell FDRE	
clocked by Clock	{rise@0.000ns fall@5.000ns period=10.000ns})	
Path Group:	Clock	
Path Type:	Hold (Min at Fast Process Corner)	
Requirement:	0.000ns (Clock rise@0.000ns - Clock	
rise@0.000ns)		
Data Path Delay:	0.345ns (logic 0.157ns (45.565%) route	

0.188ns (54.435%))

Logic Levels: 1 (LUT6=1)

Clock Path Skew: 0.031ns (DCD - SCD - CPR)

Destination Clock Delay (DCD): 1.913ns

Source Clock Delay (SCD): 1.398ns

Clock Pessimism Removal (CPR): 0.484ns

Location	Delay type	Incr(ns)
Path(ns)	Netlist Resource(s)	

	(clock Clock rise edge)	0.000
0.000 r		
D23		0.000
0.000 r	Clock (IN)	
	net (fo=0)	0.000
0.000	Clock	
D23	IBUF (Prop_ibuf_I_O)	0.155
0.155 r	Clock_IBUF_inst/O	
	net (fo=1, routed)	0.676
0.831	Clock_IBUF	
BUFGCTRL_X0Y0	BUFG (Prop_bufg_I_O)	0.026
0.857 r	Clock_IBUF_BUFG_inst/O	
	net (fo=230, routed)	0.541
1.398	Clock_IBUF_BUFG	
SLICE_X9Y78	FDRE	
r	Cycle_Count_reg[0]/C	

SLICE_X9Y78	FDRE (Prop_fdre_C_Q)	0.091
1.489 f	Cycle_Count_reg[0]/Q	
	net (fo=69, routed)	0.188
1.677	Cycle_Count_reg_n_0_[0]	
SLICE_X11Y79	LUT6 (Prop_lut6_I1_O)	0.066
1.743 r	Cycle_Count[18]_i_1/O	

	net (fo=1, routed)	0.000
1.743	Cycle_Count[18]_i_1_n_0	
	SLICE_X11Y79 FDRE	
r	Cycle_Count_reg[18]/D	

	(clock Clock rise edge)	0.000
0.000	r	
	D23	0.000
0.000	r Clock (IN)	
	net (fo=0)	0.000
0.000	Clock	
	D23 IBUF (Prop_ibuf_I_O)	0.404
0.404	r Clock_IBUF_inst/O	
	net (fo=1, routed)	0.739
1.143	Clock_IBUF	
	BUFGCTRL_X0Y0 BUFG (Prop_bufg_I_O)	0.030
1.173	r Clock_IBUF_BUFG_inst/O	
	net (fo=230, routed)	0.740
1.913	Clock_IBUF_BUFG	
	SLICE_X11Y79 FDRE	
r	Cycle_Count_reg[18]/C	
	clock pessimism	-0.484
1.429		
	SLICE_X11Y79 FDRE (Hold_fdre_C_D)	0.060
1.489	Cycle_Count_reg[18]	

	required time	
-1.489		
	arrival time	
1.743		

	slack	

0.254

Slack (MET) : 0.254ns (arrival time - required time)
Source: FSM_sequential_State_reg[0]/C
(rising edge-triggered cell FDRE
clocked by Clock {rise@0.000ns fall@5.000ns period=10.000ns})
Destination: FSM_sequential_State_reg[1]/D
(rising edge-triggered cell FDRE
clocked by Clock {rise@0.000ns fall@5.000ns period=10.000ns})
Path Group: Clock
Path Type: Hold (Min at Fast Process Corner)
Requirement: 0.000ns (Clock rise@0.000ns - Clock
rise@0.000ns)
Data Path Delay: 0.350ns (logic 0.145ns (41.389%) route
0.205ns (58.611%))
Logic Levels: 1 (LUT4=1)
Clock Path Skew: 0.000ns (DCD - SCD - CPR)
Destination Clock Delay (DCD): 1.912ns
Source Clock Delay (SCD): 1.398ns
Clock Pessimism Removal (CPR): 0.514ns

Location		Delay type	Incr(ns)
Path(ns)	Netlist Resource(s)		

		(clock Clock rise edge)	0.000
0.000 r			
D23			0.000
0.000 r	Clock (IN)		
	net (fo=0)		0.000
0.000	Clock		
D23	IBUF (Prop_ibuf_I_O)		0.155
0.155 r	Clock_IBUF_inst/O		
	net (fo=1, routed)		0.676
0.831	Clock_IBUF		
BUFGCTRL_X0Y0	BUFG (Prop_bufg_I_O)		0.026

0.857	r	Clock_IBUF_BUFG_inst/O		
		net (fo=230, routed)		0.541
1.398		Clock_IBUF_BUFG		
		SLICE_X8Y78	FDRE	
	r	FSM_sequential_State_reg[0]/C		

		SLICE_X8Y78	FDRE (Prop_fdre_C_Q)	0.118
1.516	r	FSM_sequential_State_reg[0]/Q		
		net (fo=73, routed)		0.205
1.722		State[0]		
		SLICE_X8Y78	LUT4 (Prop_lut4_I2_O)	0.027
1.749	r	FSM_sequential_State[1]_i_1/O		
		net (fo=1, routed)		0.000
1.749		FSM_sequential_State[1]_i_1_n_0		
		SLICE_X8Y78	FDRE	
	r	FSM_sequential_State_reg[1]/D		

			(clock Clock rise edge)	0.000
0.000	r			
		D23		0.000
0.000	r	Clock (IN)		
		net (fo=0)		0.000
0.000		Clock		
		D23	IBUF (Prop_ibuf_I_O)	0.404
0.404	r	Clock_IBUF_inst/O		
		net (fo=1, routed)		0.739
1.143		Clock_IBUF		
		BUFGCTRL_X0Y0	BUFG (Prop_bufg_I_O)	0.030
1.173	r	Clock_IBUF_BUFG_inst/O		
		net (fo=230, routed)		0.739
1.912		Clock_IBUF_BUFG		
		SLICE_X8Y78	FDRE	

r	FSM_sequential_State_reg[1]/C		
	clock pessimism		-0.514
1.398			
	SLICE_X8Y78	FDRE (Hold_fdre_C_D)	0.096
1.494	FSM_sequential_State_reg[1]		

	required time		
-1.494			
	arrival time		
1.749			

	slack		
0.254			

Pulse Width Checks

```
-----
-----
Clock Name:          Clock
Waveform(ns):       { 0.000 5.000 }
Period(ns):         10.000
Sources:            { Clock }

Check Type          Corner  Lib Pin          Reference Pin  Required(ns)
Actual(ns)  Slack(ns)  Location          Pin
Min Period          n/a      DSP48E1/CLK  n/a          3.292
10.000          6.708      DSP48_X0Y33
SA_Inst/Gen_PE_row[0].Gen_PE_col[1].Gen_Interior_Col.Gen_Top_Edge.P
E_Inst/MAC_Unit_Inst/Accumulator_Register_reg/CLK
Min Period          n/a      DSP48E1/CLK  n/a          3.292
10.000          6.708      DSP48_X0Y30
SA_Inst/Gen_PE_row[1].Gen_PE_col[0].Gen_Left_Edge.Gen_Left_Interior
```


.PE_Inst/MAC_Unit_Inst/Accumulator_Register_reg/CLK				
Min Period	n/a	DSP48E1/CLK	n/a	3.292
10.000	6.708	DSP48_X0Y31		
SA_Inst/Gen_PE_row[1].Gen_PE_col[1].Gen_Interior_Col.Gen_Interior.PE_Inst/MAC_Unit_Inst/Accumulator_Register_reg/CLK				
Min Period	n/a	DSP48E1/CLK	n/a	1.825
10.000	8.175	DSP48_X0Y32		
SA_Inst/Gen_PE_row[0].Gen_PE_col[0].Gen_Left_Edge.Gen_Top_Left.PE_Inst/MAC_Unit_Inst/Accumulator_Register_reg/CLK				
Min Period	n/a	BUFG/I	n/a	1.600
10.000	8.400	BUFGCTRL_X0Y0	Clock_IBUF_BUFG_inst/I	
Min Period	n/a	FDRE/C	n/a	0.750
10.000	9.250	SLICE_X11Y73	A_Reg_reg[0][0][7]/C	
Min Period	n/a	FDRE/C	n/a	0.750
10.000	9.250	SLICE_X10Y73	A_Reg_reg[0][1][4]/C	
Min Period	n/a	FDRE/C	n/a	0.750
10.000	9.250	SLICE_X10Y73	A_Reg_reg[0][1][7]/C	
Min Period	n/a	FDRE/C	n/a	0.750
10.000	9.250	SLICE_X13Y78	B_Reg_reg[1][0][4]/C	
Min Period	n/a	FDRE/C	n/a	0.750
10.000	9.250	SLICE_X13Y78	B_Reg_reg[1][0][7]/C	
Low Pulse Width	Slow	FDRE/C	n/a	0.400
5.000	4.600	SLICE_X11Y73	A_Reg_reg[0][0][7]/C	
Low Pulse Width	Fast	FDRE/C	n/a	0.400
5.000	4.600	SLICE_X11Y73	A_Reg_reg[0][0][7]/C	
Low Pulse Width	Slow	FDRE/C	n/a	0.400
5.000	4.600	SLICE_X10Y73	A_Reg_reg[0][1][4]/C	
Low Pulse Width	Fast	FDRE/C	n/a	0.400
5.000	4.600	SLICE_X10Y73	A_Reg_reg[0][1][4]/C	
Low Pulse Width	Slow	FDRE/C	n/a	0.400
5.000	4.600	SLICE_X10Y73	A_Reg_reg[0][1][7]/C	
Low Pulse Width	Fast	FDRE/C	n/a	0.400
5.000	4.600	SLICE_X10Y73	A_Reg_reg[0][1][7]/C	
Low Pulse Width	Slow	FDRE/C	n/a	0.400
5.000	4.600	SLICE_X13Y78	B_Reg_reg[1][0][4]/C	
Low Pulse Width	Fast	FDRE/C	n/a	0.400
5.000	4.600	SLICE_X13Y78	B_Reg_reg[1][0][4]/C	

Low Pulse Width	Slow	FDRE/C	n/a	0.400
5.000	4.600	SLICE_X13Y78	B_Reg_reg[1][0][7]/C	
Low Pulse Width	Fast	FDRE/C	n/a	0.400
5.000	4.600	SLICE_X13Y78	B_Reg_reg[1][0][7]/C	
High Pulse Width	Slow	FDRE/C	n/a	0.350
5.000	4.650	SLICE_X11Y73	A_Reg_reg[0][0][0]/C	
High Pulse Width	Fast	FDRE/C	n/a	0.350
5.000	4.650	SLICE_X11Y73	A_Reg_reg[0][0][0]/C	
High Pulse Width	Slow	FDRE/C	n/a	0.350
5.000	4.650	SLICE_X11Y73	A_Reg_reg[0][0][1]/C	
High Pulse Width	Fast	FDRE/C	n/a	0.350
5.000	4.650	SLICE_X11Y73	A_Reg_reg[0][0][1]/C	
High Pulse Width	Slow	FDRE/C	n/a	0.350
5.000	4.650	SLICE_X11Y73	A_Reg_reg[0][0][2]/C	
High Pulse Width	Fast	FDRE/C	n/a	0.350
5.000	4.650	SLICE_X11Y73	A_Reg_reg[0][0][2]/C	
High Pulse Width	Slow	FDRE/C	n/a	0.350
5.000	4.650	SLICE_X11Y73	A_Reg_reg[0][0][3]/C	
High Pulse Width	Fast	FDRE/C	n/a	0.350
5.000	4.650	SLICE_X11Y73	A_Reg_reg[0][0][3]/C	
High Pulse Width	Slow	FDRE/C	n/a	0.350
5.000	4.650	SLICE_X11Y74	A_Reg_reg[0][0][4]/C	
High Pulse Width	Fast	FDRE/C	n/a	0.350
5.000	4.650	SLICE_X11Y74	A_Reg_reg[0][0][4]/C	

Path Group:	(none)			
From Clock:	Clock			
To Clock:				
Max Delay	129 Endpoints			
Min Delay	129 Endpoints			

Max Delay Paths

Slack: inf
Source: Output_Reg_reg[1][1][31]/C
(rising edge-triggered cell FDRE
clocked by Clock {rise@0.000ns fall@5.000ns period=10.000ns})
Destination: Output[1][1][31]
(output port)
Path Group: (none)
Path Type: Max at Slow Process Corner
Data Path Delay: 7.532ns (logic 2.772ns (36.808%) route
4.759ns (63.192%))
Logic Levels: 1 (OBUF=1)
Clock Uncertainty: 0.025ns ((TSJ^2 + TIJ^2)^1/2 + DJ) / 2 +
PE

Total System Jitter	(TSJ):	0.050ns
Total Input Jitter	(TIJ):	0.000ns
Discrete Jitter	(DJ):	0.000ns
Phase Error	(PE):	0.000ns

Location	Delay type	Incr(ns)
Path(ns)	Netlist Resource(s)	

	(clock Clock rise edge)	0.000
0.000 r		
D23		0.000
0.000 r	Clock (IN)	
	net (fo=0)	0.000
0.000	Clock	
D23	IBUF (Prop_ibuf_I_O)	0.892
0.892 r	Clock_IBUF_inst/O	
	net (fo=1, routed)	1.764

```

2.656      Clock_IBUF
          BUFGCTRL_X0Y0          BUFG (Prop_bufg_I_O)          0.120
2.776 r    Clock_IBUF_BUFG_inst/O
          net (fo=230, routed)          1.385
4.161      Clock_IBUF_BUFG
          SLICE_X13Y77          FDRE
r    Output_Reg_reg[1][1][31]/C
-----
          SLICE_X13Y77          FDRE (Prop_fdre_C_Q)          0.269
4.430 r    Output_Reg_reg[1][1][31]/Q
          net (fo=1, routed)          4.759
9.189      Output[1][1]_OBUF[31]
          A8          OBUF (Prop_obuf_I_O)          2.503
11.693 r    Output[1][1][31]_INST_0/O
          net (fo=0)          0.000
11.693      Output[1][1][31]
          A8
r    Output[1][1][31] (OUT)
-----
-----

Slack:          inf
Source:          Output_Reg_reg[1][1][27]/C
                  (rising edge-triggered cell FDRE
clocked by Clock {rise@0.000ns fall@5.000ns period=10.000ns})
Destination:     Output[1][1][27]
                  (output port)

Path Group:      (none)
Path Type:        Max at Slow Process Corner
Data Path Delay:  7.203ns  (logic 2.861ns (39.723%)  route
4.342ns (60.277%))
Logic Levels:     1  (OBUF=1)
Clock Uncertainty: 0.025ns  ((TSJ^2 + TIJ^2)^1/2 + DJ) / 2 +
PE

```

Total System Jitter	(TSJ):	0.050ns
Total Input Jitter	(TIJ):	0.000ns
Discrete Jitter	(DJ):	0.000ns
Phase Error	(PE):	0.000ns

Location	Delay type	Incr(ns)
Path(ns)	Netlist Resource(s)	

```

-----
                                (clock Clock rise edge)          0.000
0.000 r
      D23                                0.000
0.000 r  Clock (IN)
                                net (fo=0)                    0.000
0.000    Clock
      D23                                IBUF (Prop_ibuf_I_O)      0.892
0.892 r  Clock_IBUF_inst/O
                                net (fo=1, routed)             1.764
2.656    Clock_IBUF
      BUFGCTRL_X0Y0                    BUFG (Prop_bufg_I_O)      0.120
2.776 r  Clock_IBUF_BUFG_inst/O
                                net (fo=230, routed)           1.388
4.164    Clock_IBUF_BUFG
      SLICE_X8Y79                      FDRE
r  Output_Reg_reg[1][1][27]/C

```

```

-----
      SLICE_X8Y79                      FDRE (Prop_fdre_C_Q)      0.282
4.446 r  Output_Reg_reg[1][1][27]/Q
                                net (fo=1, routed)             4.342
8.788    Output[1][1]_OBUF[27]
      F10                                OBUF (Prop_obuf_I_O)      2.579
11.367 r  Output[1][1][27]_INST_0/O
                                net (fo=0)                    0.000
11.367    Output[1][1][27]

```

```

F10
r  Output[1][1][27]  (OUT)
-----
-----

Slack:                                inf
Source:                              Output_Reg_reg[1][1][29]/C
                                     (rising edge-triggered cell FDRE
clocked by Clock  {rise@0.000ns fall@5.000ns period=10.000ns})
Destination:                          Output[1][1][29]
                                     (output port)

Path Group:                           (none)
Path Type:                            Max at Slow Process Corner
Data Path Delay:                       7.147ns  (logic 2.773ns (38.806%)  route
4.373ns (61.194%))
Logic Levels:                          1  (OBUF=1)
Clock Uncertainty:                     0.025ns  ((TSJ^2 + TIJ^2)^1/2 + DJ) / 2 +
PE

Total System Jitter      (TSJ):      0.050ns
Total Input Jitter       (TIJ):      0.000ns
Discrete Jitter          (DJ):       0.000ns
Phase Error              (PE):       0.000ns


Location                    Delay type                    Incr(ns)
Path(ns)    Netlist Resource(s)
-----
-----

                                (clock Clock rise edge)      0.000
0.000 r
    D23                                0.000
0.000 r  Clock  (IN)
                                net  (fo=0)                  0.000
0.000    Clock
    D23                                IBUF  (Prop_ibuf_I_O)      0.892
0.892 r  Clock_IBUF_inst/O

```

```

                                net (fo=1, routed)                                1.764
2.656      Clock_IBUF
      BUFGCTRL_X0Y0      BUFG (Prop_bufg_I_O)                                0.120
2.776 r   Clock_IBUF_BUFG_inst/O
                                net (fo=230, routed)                            1.389
4.165      Clock_IBUF_BUFG
      SLICE_X9Y80      FDRE
r   Output_Reg_reg[1][1][29]/C
-----
-----
      SLICE_X9Y80      FDRE (Prop_fdre_C_Q)                                0.269
4.434 r   Output_Reg_reg[1][1][29]/Q
                                net (fo=1, routed)                            4.373
8.807      Output[1][1]_OBUF[29]
      B9      OBUF (Prop_obuf_I_O)                                2.504
11.311 r   Output[1][1][29]_INST_0/O
                                net (fo=0)                                    0.000
11.311      Output[1][1][29]
      B9
r   Output[1][1][29] (OUT)
-----
-----

```

```

Slack:                                inf
Source:                                Output_Reg_reg[1][1][20]/C
                                         (rising edge-triggered cell FDRE
clocked by Clock {rise@0.000ns fall@5.000ns period=10.000ns})
Destination:                           Output[1][1][20]
                                         (output port)
Path Group:                             (none)
Path Type:                              Max at Slow Process Corner
Data Path Delay:                         7.116ns  (logic 2.861ns (40.204%)  route
4.255ns (59.796%))
Logic Levels:                           1  (OBUF=1)
Clock Uncertainty:                       0.025ns  ((TSJ^2 + TIJ^2)^1/2 + DJ) / 2 +

```

PE

Total System Jitter	(TSJ):	0.050ns
Total Input Jitter	(TIJ):	0.000ns
Discrete Jitter	(DJ):	0.000ns
Phase Error	(PE):	0.000ns

Location	Delay type	Incr(ns)
Path(ns)	Netlist Resource(s)	

		(clock Clock rise edge)	0.000
0.000 r			
D23			0.000
0.000 r	Clock (IN)		
		net (fo=0)	0.000
0.000	Clock		
D23		IBUF (Prop_ibuf_I_O)	0.892
0.892 r	Clock_IBUF_inst/O		
		net (fo=1, routed)	1.764
2.656	Clock_IBUF		
	BUFGCTRL_X0Y0	BUFG (Prop_bufg_I_O)	0.120
2.776 r	Clock_IBUF_BUFG_inst/O		
		net (fo=230, routed)	1.385
4.161	Clock_IBUF_BUFG		
	SLICE_X8Y77	FDRE	
r	Output_Reg_reg[1][1][20]/C		

	SLICE_X8Y77	FDRE (Prop_fdre_C_Q)	0.282
4.443 r	Output_Reg_reg[1][1][20]/Q		
		net (fo=1, routed)	4.255
8.698	Output[1][1]_OBUF[20]		
F14		OBUF (Prop_obuf_I_O)	2.579
11.277 r	Output[1][1][20]_INST_0/O		
		net (fo=0)	0.000


```
11.277      Output[1][1][20]
      F14
r  Output[1][1][20]  (OUT)

-----

Slack:                inf
Source:               Output_Reg_reg[1][1][22]/C
                     (rising edge-triggered cell FDRE
clocked by Clock  {rise@0.000ns fall@5.000ns period=10.000ns})
Destination:         Output[1][1][22]
                     (output port)

Path Group:          (none)
Path Type:           Max at Slow Process Corner
Data Path Delay:     7.098ns  (logic 2.769ns (39.007%)  route
4.329ns (60.993%))
Logic Levels:       1  (OBUF=1)
Clock Uncertainty:   0.025ns  ((TSJ^2 + TIJ^2)^1/2 + DJ) / 2 +
PE

Total System Jitter    (TSJ):    0.050ns
Total Input Jitter     (TIJ):    0.000ns
Discrete Jitter        (DJ):    0.000ns
Phase Error            (PE):    0.000ns


Location              Delay type              Incr(ns)
Path(ns)      Netlist Resource(s)

-----

0.000 r              (clock Clock rise edge)      0.000
      D23
0.000 r  Clock (IN)              0.000
      net (fo=0)
0.000      Clock              0.000
      D23      IBUF (Prop_ibuf_I_O)      0.892
```

```

0.892 r  Clock_IBUF_inst/O
                                net (fo=1, routed)                1.764
2.656    Clock_IBUF
    BUFGCTRL_X0Y0    BUFG (Prop_bufg_I_O)                0.120
2.776 r  Clock_IBUF_BUFG_inst/O
                                net (fo=230, routed)            1.385
4.161    Clock_IBUF_BUFG
    SLICE_X10Y77    FDRE
r  Output_Reg_reg[1][1][22]/C
-----
-----
    SLICE_X10Y77    FDRE (Prop_fdre_C_Q)                0.308
4.469 r  Output_Reg_reg[1][1][22]/Q
                                net (fo=1, routed)            4.329
8.798    Output[1][1]_OBUF[22]
    E11    OBUF (Prop_obuf_I_O)                2.461
11.259 r  Output[1][1][22]_INST_0/O
                                net (fo=0)                    0.000
11.259    Output[1][1][22]
    E11
r  Output[1][1][22] (OUT)
-----
-----

Slack:                inf
Source:                Output_Reg_reg[1][1][23]/C
                      (rising edge-triggered cell FDRE
clocked by Clock {rise@0.000ns fall@5.000ns period=10.000ns})
Destination:          Output[1][1][23]
                      (output port)
Path Group:            (none)
Path Type:             Max at Slow Process Corner
Data Path Delay:       7.081ns (logic 2.871ns (40.539%) route
4.210ns (59.461%))
Logic Levels:          1 (OBUF=1)

```

Clock Uncertainty: 0.025ns ((TSJ^2 + TIJ^2)^1/2 + DJ) / 2 + PE

Total System Jitter	(TSJ):	0.050ns
Total Input Jitter	(TIJ):	0.000ns
Discrete Jitter	(DJ):	0.000ns
Phase Error	(PE):	0.000ns

Location	Delay type	Incr(ns)
Path(ns) Netlist Resource(s)		

	(clock Clock rise edge)	0.000
0.000 r		
D23		0.000
0.000 r	Clock (IN)	
	net (fo=0)	0.000
0.000	Clock	
D23	IBUF (Prop_ibuf_I_O)	0.892
0.892 r	Clock_IBUF_inst/O	
	net (fo=1, routed)	1.764
2.656	Clock_IBUF	
BUFGCTRL_X0Y0	BUFG (Prop_bufg_I_O)	0.120
2.776 r	Clock_IBUF_BUFG_inst/O	
	net (fo=230, routed)	1.385
4.161	Clock_IBUF_BUFG	
SLICE_X8Y77	FDRE	
r	Output_Reg_reg[1][1][23]/C	

SLICE_X8Y77	FDRE (Prop_fdre_C_Q)	0.282
4.443 r	Output_Reg_reg[1][1][23]/Q	
	net (fo=1, routed)	4.210
8.653	Output[1][1]_OBUF[23]	
C11	OBUF (Prop_obuf_I_O)	2.589
11.242 r	Output[1][1][23]_INST_0/O	

```

net (fo=0) 0.000
11.242 Output[1][1][23]
C11
r Output[1][1][23] (OUT)

-----

Slack: inf
Source: Output_Reg_reg[1][1][24]/C
        (rising edge-triggered cell FDRE
clocked by Clock {rise@0.000ns fall@5.000ns period=10.000ns})
Destination: Output[1][1][24]
              (output port)
Path Group: (none)
Path Type: Max at Slow Process Corner
Data Path Delay: 6.997ns (logic 2.864ns (40.934%) route
4.133ns (59.066%))
Logic Levels: 1 (OBUF=1)
Clock Uncertainty: 0.025ns ((TSJ^2 + TIJ^2)^1/2 + DJ) / 2 +
PE
Total System Jitter (TSJ): 0.050ns
Total Input Jitter (TIJ): 0.000ns
Discrete Jitter (DJ): 0.000ns
Phase Error (PE): 0.000ns

Location Delay type Incr(ns)
Path(ns) Netlist Resource(s)
-----
0.000 r (clock Clock rise edge) 0.000
D23 0.000
0.000 r Clock (IN) 0.000
net (fo=0) 0.000
0.000 Clock

```

D23	IBUF (Prop_ibuf_I_O)	0.892
0.892 r	Clock_IBUF_inst/O	
	net (fo=1, routed)	1.764
2.656	Clock_IBUF	
	BUFGCTRL_X0Y0	0.120
2.776 r	Clock_IBUF_BUFG_inst/O	
	net (fo=230, routed)	1.388
4.164	Clock_IBUF_BUFG	
	SLICE_X8Y79	FDRE
r	Output_Reg_reg[1][1][24]/C	

	SLICE_X8Y79	FDRE (Prop_fdre_C_Q)	0.282
4.446 r	Output_Reg_reg[1][1][24]/Q		
	net (fo=1, routed)		4.133
8.579	Output[1][1]_OBUF[24]		
	C12	OBUF (Prop_obuf_I_O)	2.582
11.161 r	Output[1][1][24]_INST_0/O		
	net (fo=0)		0.000
11.161	Output[1][1][24]		
	C12		
r	Output[1][1][24] (OUT)		

Slack:	inf
Source:	Output_Reg_reg[1][1][15]/C
	(rising edge-triggered cell FDRE
clocked by Clock	{rise@0.000ns fall@5.000ns period=10.000ns})
Destination:	Output[1][1][15]
	(output port)
Path Group:	(none)
Path Type:	Max at Slow Process Corner
Data Path Delay:	6.960ns (logic 2.820ns (40.512%) route
4.140ns (59.488%))	

Logic Levels:	1	(OBUF=1)	
Clock Uncertainty:	0.025ns	$((TSJ^2 + TIJ^2)^{1/2} + DJ) / 2 + PE$	
Total System Jitter	(TSJ):	0.050ns	
Total Input Jitter	(TIJ):	0.000ns	
Discrete Jitter	(DJ):	0.000ns	
Phase Error	(PE):	0.000ns	
Location	Delay type	Incr(ns)	
Path(ns)	Netlist Resource(s)		

	(clock Clock rise edge)	0.000	
0.000 r			
D23		0.000	
0.000 r	Clock (IN)		
	net (fo=0)	0.000	
0.000	Clock		
D23	IBUF (Prop_ibuf_I_O)	0.892	
0.892 r	Clock_IBUF_inst/O		
	net (fo=1, routed)	1.764	
2.656	Clock_IBUF		
BUFGCTRL_X0Y0	BUFG (Prop_bufg_I_O)	0.120	
2.776 r	Clock_IBUF_BUFG_inst/O		
	net (fo=230, routed)	1.388	
4.164	Clock_IBUF_BUFG		
SLICE_X9Y79	FDRE		
r	Output_Reg_reg[1][1][15]/C		

	SLICE_X9Y79	FDRE (Prop_fdre_C_Q)	0.246
4.410 r	Output_Reg_reg[1][1][15]/Q		
	net (fo=1, routed)	4.140	
8.550	Output[1][1]_OBUF[15]		
D13	OBUF (Prop_obuf_I_O)	2.574	

```

11.124 r   Output[1][1][15]_INST_0/O
                                net (fo=0)                                0.000

11.124      Output[1][1][15]
      D13
r   Output[1][1][15] (OUT)

-----
-----

Slack:                                inf
Source:                                Output_Reg_reg[1][1][14]/C
                                         (rising edge-triggered cell FDRE
clocked by Clock {rise@0.000ns fall@5.000ns period=10.000ns})
Destination:                           Output[1][1][14]
                                         (output port)

Path Group:                             (none)
Path Type:                              Max at Slow Process Corner
Data Path Delay:                         6.876ns  (logic 2.760ns (40.144%)  route
4.116ns (59.856%))
Logic Levels:                           1  (OBUF=1)
Clock Uncertainty:                       0.025ns  ((TSJ^2 + TIJ^2)^1/2 + DJ) / 2 +
PE

Total System Jitter      (TSJ):      0.050ns
Total Input Jitter       (TIJ):      0.000ns
Discrete Jitter          (DJ):      0.000ns
Phase Error              (PE):      0.000ns


Location                Delay type                Incr(ns)
Path(ns)      Netlist Resource(s)

-----
-----

                                (clock Clock rise edge)                0.000
0.000 r
      D23                                0.000
0.000 r   Clock (IN)
                                net (fo=0)                                0.000

```

0.000	Clock		
	D23	IBUF (Prop_ibuf_I_O)	0.892
0.892	r Clock_IBUF_inst/O		
		net (fo=1, routed)	1.764
2.656	Clock_IBUF		
	BUFGCTRL_X0Y0	BUFG (Prop_bufg_I_O)	0.120
2.776	r Clock_IBUF_BUFG_inst/O		
		net (fo=230, routed)	1.388
4.164	Clock_IBUF_BUFG		
	SLICE_X10Y79	FDRE	
r	Output_Reg_reg[1][1][14]/C		

	SLICE_X10Y79	FDRE (Prop_fdre_C_Q)	0.308
4.472	r Output_Reg_reg[1][1][14]/Q		
		net (fo=1, routed)	4.116
8.588	Output[1][1]_OBUF[14]		
	E13	OBUF (Prop_obuf_I_O)	2.452
11.040	r Output[1][1][14]_INST_0/O		
		net (fo=0)	0.000
11.040	Output[1][1][14]		
	E13		
r	Output[1][1][14] (OUT)		

Slack: inf

Source: Output_Reg_reg[1][1][17]/C
(rising edge-triggered cell FDRE)

clocked by Clock {rise@0.000ns fall@5.000ns period=10.000ns})

Destination: Output[1][1][17]
(output port)

Path Group: (none)

Path Type: Max at Slow Process Corner

Data Path Delay: 6.808ns (logic 2.801ns (41.137%) route)

4.008ns (58.863%))

Logic Levels:1 (OBUF=1)

Clock Uncertainty:0.025ns ((TSJ^2 + TIJ^2)^1/2 + DJ) / 2 + PE

Total System Jitter(TSJ):0.050ns

Total Input Jitter(TIJ):0.000ns

Discrete Jitter(DJ):0.000ns

Phase Error(PE):0.000ns

Location	Delay type	Incr(ns)
Path(ns)	Netlist Resource(s)	

	(clock Clock rise edge)	0.000
0.000 r		
D23		0.000
0.000 r	Clock (IN)	
	net (fo=0)	0.000
0.000	Clock	
D23	IBUF (Prop_ibuf_I_O)	0.892
0.892 r	Clock_IBUF_inst/O	
	net (fo=1, routed)	1.764
2.656	Clock_IBUF	
BUFGCTRL_X0Y0	BUFG (Prop_bufg_I_O)	0.120
2.776 r	Clock_IBUF_BUFG_inst/O	
	net (fo=230, routed)	1.388
4.164	Clock_IBUF_BUFG	
SLICE_X9Y79	FDRE	
r	Output_Reg_reg[1][1][17]/C	

SLICE_X9Y79	FDRE (Prop_fdre_C_Q)	0.246
4.410 r	Output_Reg_reg[1][1][17]/Q	
	net (fo=1, routed)	4.008
8.417	Output[1][1]_OBUF[17]	

```

F12                                OBUF (Prop_obuf_I_O)                2.555
10.972 r  Output[1][1][17]_INST_0/O
                                net (fo=0)                        0.000
10.972      Output[1][1][17]
F12
r  Output[1][1][17] (OUT)

-----
-----

Min Delay Paths
-----
-----

Slack:                            inf
Source:                            Output_Reg_reg[0][0][15]/C
                                (rising edge-triggered cell FDRE
clocked by Clock {rise@0.000ns fall@5.000ns period=10.000ns})
Destination:                       Output[0][0][15]
                                (output port)

Path Group:                         (none)
Path Type:                          Min at Fast Process Corner
Data Path Delay:                    1.854ns  (logic 1.440ns (77.689%)  route
0.414ns (22.311%))
Logic Levels:                       1  (OBUF=1)
Clock Uncertainty:                  0.025ns  ((TSJ^2 + TIJ^2)^1/2 + DJ) / 2 +
PE

Total System Jitter      (TSJ):      0.050ns
Total Input Jitter       (TIJ):      0.000ns
Discrete Jitter          (DJ):       0.000ns
Phase Error              (PE):       0.000ns


Location                  Delay type                Incr(ns)
Path(ns)      Netlist Resource(s)

```

```

-----
                                (clock Clock rise edge)          0.000
0.000 r
    D23                                0.000
0.000 r  Clock (IN)
                                net (fo=0)                    0.000
0.000    Clock
    D23                                IBUF (Prop_ibuf_I_O)      0.155
0.155 r  Clock_IBUF_inst/O
                                net (fo=1, routed)            0.676
0.831    Clock_IBUF
    BUFGCTRL_X0Y0                    BUFG (Prop_bufg_I_O)      0.026
0.857 r  Clock_IBUF_BUFG_inst/O
                                net (fo=230, routed)          0.546
1.403    Clock_IBUF_BUFG
    SLICE_X11Y83                      FDRE
r  Output_Reg_reg[0][0][15]/C

```

```

-----
    SLICE_X11Y83                      FDRE (Prop_fdre_C_Q)      0.100
1.503 r  Output_Reg_reg[0][0][15]/Q
                                net (fo=1, routed)            0.414
1.917    Output[0][0]_OBUF[15]
    E21                                OBUF (Prop_obuf_I_O)      1.340
3.257 r  Output[0][0][15]_INST_0/O
                                net (fo=0)                    0.000
3.257    Output[0][0][15]
    E21
r  Output[0][0][15] (OUT)

```

```

-----
Slack:                                inf

```

```

Source:                                Output_Reg_reg[0][0][16]/C
                                      (rising edge-triggered cell FDRE
clocked by Clock {rise@0.000ns fall@5.000ns period=10.000ns})
Destination:                          Output[0][0][16]
                                      (output port)

Path Group:                           (none)
Path Type:                            Min at Fast Process Corner
Data Path Delay:                      1.873ns  (logic 1.463ns (78.091%)  route
0.410ns (21.909%))
Logic Levels:                        1  (OBUF=1)
Clock Uncertainty:                   0.025ns  ((TSJ^2 + TIJ^2)^1/2 + DJ) / 2 +
PE

```

```

Total System Jitter      (TSJ):      0.050ns
Total Input Jitter       (TIJ):      0.000ns
Discrete Jitter          (DJ):       0.000ns
Phase Error              (PE):       0.000ns

```

Location	Delay type	Incr(ns)
Path(ns) Netlist Resource(s)		

	(clock Clock rise edge)	0.000
0.000 r		
D23		0.000
0.000 r Clock (IN)		
	net (fo=0)	0.000
0.000 Clock		
D23	IBUF (Prop_ibuf_I_O)	0.155
0.155 r Clock_IBUF_inst/O		
	net (fo=1, routed)	0.676
0.831 Clock_IBUF		
BUFGCTRL_X0Y0	BUFG (Prop_bufg_I_O)	0.026
0.857 r Clock_IBUF_BUFG_inst/O		
	net (fo=230, routed)	0.546
1.403 Clock_IBUF_BUFG		
SLICE_X11Y83	FDRE	

```

r   Output_Reg_reg[0][0][16]/C
-----
-----
      SLICE_X11Y83          FDRE  (Prop_fdre_C_Q)          0.100
1.503 r   Output_Reg_reg[0][0][16]/Q
              net (fo=1, routed)          0.410
1.914      Output[0][0]_OBUF[16]
      A20          OBUF  (Prop_obuf_I_O)          1.363
3.276 r   Output[0][0][16]_INST_0/O
              net (fo=0)          0.000
3.276      Output[0][0][16]
      A20
r   Output[0][0][16] (OUT)
-----
-----

Slack:          inf
Source:          Output_Reg_reg[0][0][17]/C
                  (rising edge-triggered cell FDRE
clocked by Clock {rise@0.000ns fall@5.000ns period=10.000ns})
Destination:     Output[0][0][17]
                  (output port)

Path Group:      (none)
Path Type:       Min at Fast Process Corner
Data Path Delay: 1.879ns  (logic 1.472ns (78.350%)  route
0.407ns (21.650%))
Logic Levels:    1  (OBUF=1)
Clock Uncertainty: 0.025ns  ((TSJ^2 + TIJ^2)^1/2 + DJ) / 2 +
PE

Total System Jitter      (TSJ):      0.050ns
Total Input Jitter       (TIJ):      0.000ns
Discrete Jitter          (DJ):      0.000ns
Phase Error              (PE):      0.000ns

Location                Delay type                Incr(ns)

```

Path(ns)	Netlist Resource(s)		

		(clock Clock rise edge)	0.000
0.000 r			
	D23		0.000
0.000 r	Clock (IN)		
		net (fo=0)	0.000
0.000	Clock		
	D23	IBUF (Prop_ibuf_I_O)	0.155
0.155 r	Clock_IBUF_inst/O		
		net (fo=1, routed)	0.676
0.831	Clock_IBUF		
	BUFGCTRL_X0Y0	BUFG (Prop_bufg_I_O)	0.026
0.857 r	Clock_IBUF_BUFG_inst/O		
		net (fo=230, routed)	0.547
1.404	Clock_IBUF_BUFG		
	SLICE_X10Y84	FDRE	
r	Output_Reg_reg[0][0][17]/C		

	SLICE_X10Y84	FDRE (Prop_fdre_C_Q)	0.118
1.522 r	Output_Reg_reg[0][0][17]/Q		
		net (fo=1, routed)	0.407
1.929	Output[0][0]_OBUF[17]		
	B20	OBUF (Prop_obuf_I_O)	1.354
3.284 r	Output[0][0][17]_INST_0/O		
		net (fo=0)	0.000
3.284	Output[0][0][17]		
	B20		
r	Output[0][0][17] (OUT)		


```

Slack:                                inf
Source:                               Output_Reg_reg[0][0][12]/C
                                      (rising edge-triggered cell FDRE
clocked by Clock {rise@0.000ns fall@5.000ns period=10.000ns})
Destination:                          Output[0][0][12]
                                      (output port)

Path Group:                           (none)
Path Type:                            Min at Fast Process Corner
Data Path Delay:                      1.886ns  (logic 1.458ns (77.316%)  route
0.428ns (22.684%))
Logic Levels:                        1  (OBUF=1)
Clock Uncertainty:                    0.025ns  ((TSJ^2 + TIJ^2)^1/2 + DJ) / 2 +
PE
Total System Jitter      (TSJ):      0.050ns
Total Input Jitter       (TIJ):      0.000ns
Discrete Jitter          (DJ):      0.000ns
Phase Error              (PE):      0.000ns

```

Location	Delay type	Incr(ns)
Path(ns) Netlist Resource(s)		

	(clock Clock rise edge)	0.000
0.000 r		
D23		0.000
0.000 r Clock (IN)		
	net (fo=0)	0.000
0.000 Clock		
D23	IBUF (Prop_ibuf_I_O)	0.155
0.155 r Clock_IBUF_inst/O		
	net (fo=1, routed)	0.676
0.831 Clock_IBUF		
BUFGCTRL_X0Y0	BUFG (Prop_bufg_I_O)	0.026
0.857 r Clock_IBUF_BUFG_inst/O		
	net (fo=230, routed)	0.546
1.403 Clock_IBUF_BUFG		

```

SLICE_X11Y83          FDRE
r  Output_Reg_reg[0][0][12]/C

-----

SLICE_X11Y83          FDRE  (Prop_fdre_C_Q)          0.100
1.503 r  Output_Reg_reg[0][0][12]/Q
          net  (fo=1, routed)          0.428
1.931    Output[0][0]_OBUF[12]
          B21          OBUF  (Prop_obuf_I_O)          1.358
3.289 r  Output[0][0][12]_INST_0/O
          net  (fo=0)          0.000
3.289    Output[0][0][12]
          B21
r  Output[0][0][12]  (OUT)

-----

Slack:                inf
Source:                Output_Reg_reg[0][0][13]/C
                      (rising edge-triggered cell FDRE
clocked by Clock  {rise@0.000ns fall@5.000ns period=10.000ns})
Destination:          Output[0][0][13]
                      (output port)

Path Group:            (none)
Path Type:             Min at Fast Process Corner
Data Path Delay:       1.890ns  (logic 1.446ns (76.500%)  route
0.444ns (23.500%))
Logic Levels:          1  (OBUF=1)
Clock Uncertainty:     0.025ns  ((TSJ^2 + TIJ^2)^1/2 + DJ) / 2 +
PE

Total System Jitter    (TSJ):      0.050ns
Total Input Jitter     (TIJ):      0.000ns
Discrete Jitter        (DJ):      0.000ns
Phase Error            (PE):      0.000ns

```


Location	Delay	type	Incr(ns)
Path(ns)	Netlist	Resource(s)	

		(clock Clock rise edge)	0.000
0.000 r			
	D23		0.000
0.000 r	Clock (IN)		
		net (fo=0)	0.000
0.000	Clock		
	D23	IBUF (Prop_ibuf_I_O)	0.155
0.155 r	Clock_IBUF_inst/O		
		net (fo=1, routed)	0.676
0.831	Clock_IBUF		
	BUFGCTRL_X0Y0	BUFG (Prop_bufg_I_O)	0.026
0.857 r	Clock_IBUF_BUFG_inst/O		
		net (fo=230, routed)	0.545
1.402	Clock_IBUF_BUFG		
	SLICE_X9Y82	FDRE	
r	Output_Reg_reg[0][0][13]/C		

	SLICE_X9Y82	FDRE (Prop_fdre_C_Q)	0.100
1.502 r	Output_Reg_reg[0][0][13]/Q		
		net (fo=1, routed)	0.444
1.946	Output[0][0]_OBUF[13]		
	C21	OBUF (Prop_obuf_I_O)	1.346
3.292 r	Output[0][0][13]_INST_0/O		
		net (fo=0)	0.000
3.292	Output[0][0][13]		
	C21		
r	Output[0][0][13] (OUT)		

Slack: inf

Source: Output_Reg_reg[0][0][14]/C
(rising edge-triggered cell FDRE
clocked by Clock {rise@0.000ns fall@5.000ns period=10.000ns})

Destination: Output[0][0][14]
(output port)

Path Group: (none)

Path Type: Min at Fast Process Corner

Data Path Delay: 1.896ns (logic 1.441ns (75.998%) route
0.455ns (24.002%))

Logic Levels: 1 (OBUF=1)

Clock Uncertainty: 0.025ns ((TSJ^2 + TIJ^2)^1/2 + DJ) / 2 +
PE

Total System Jitter	(TSJ):	0.050ns
Total Input Jitter	(TIJ):	0.000ns
Discrete Jitter	(DJ):	0.000ns
Phase Error	(PE):	0.000ns

Location	Delay type	Incr(ns)
Path(ns) Netlist Resource(s)		

	(clock Clock rise edge)	0.000
0.000 r		
D23		0.000
0.000 r Clock (IN)		
	net (fo=0)	0.000
0.000 Clock		
D23	IBUF (Prop_ibuf_I_O)	0.155
0.155 r Clock_IBUF_inst/O		
	net (fo=1, routed)	0.676
0.831 Clock_IBUF		
BUFGCTRL_X0Y0	BUFG (Prop_bufg_I_O)	0.026
0.857 r Clock_IBUF_BUFG_inst/O		
	net (fo=230, routed)	0.546

```

1.403      Clock_IBUF_BUFG
          SLICE_X11Y83          FDRE
r  Output_Reg_reg[0][0][14]/C

-----

          SLICE_X11Y83          FDRE (Prop_fdre_C_Q)          0.100
1.503 r  Output_Reg_reg[0][0][14]/Q
          net (fo=1, routed)          0.455
1.958      Output[0][0]_OBUF[14]
          E22          OBUF (Prop_obuf_I_O)          1.341
3.299 r  Output[0][0][14]_INST_0/O
          net (fo=0)          0.000
3.299      Output[0][0][14]
          E22
r  Output[0][0][14] (OUT)

-----

Slack:          inf
Source:          Output_Reg_reg[0][0][18]/C
                  (rising edge-triggered cell FDRE
clocked by Clock {rise@0.000ns fall@5.000ns period=10.000ns})
Destination:     Output[0][0][18]
                  (output port)
Path Group:      (none)
Path Type:       Min at Fast Process Corner
Data Path Delay: 1.896ns  (logic 1.483ns (78.188%)  route
0.414ns (21.812%))
Logic Levels:    1  (OBUF=1)
Clock Uncertainty: 0.025ns  ((TSJ^2 + TIJ^2)^1/2 + DJ) / 2 +
PE

Total System Jitter    (TSJ):    0.050ns
Total Input Jitter     (TIJ):    0.000ns
Discrete Jitter        (DJ):    0.000ns
Phase Error            (PE):    0.000ns

```

Location	Delay type	Incr(ns)
Path(ns)	Netlist Resource(s)	

	(clock Clock rise edge)	0.000
0.000 r		
D23		0.000
0.000 r	Clock (IN)	
	net (fo=0)	0.000
0.000	Clock	
D23	IBUF (Prop_ibuf_I_O)	0.155
0.155 r	Clock_IBUF_inst/O	
	net (fo=1, routed)	0.676
0.831	Clock_IBUF	
BUFGCTRL_X0Y0	BUFG (Prop_bufg_I_O)	0.026
0.857 r	Clock_IBUF_BUFG_inst/O	
	net (fo=230, routed)	0.547
1.404	Clock_IBUF_BUFG	
SLICE_X10Y84	FDRE	
r	Output_Reg_reg[0][0][18]/C	

	SLICE_X10Y84	FDRE (Prop_fdre_C_Q)
1.522 r	Output_Reg_reg[0][0][18]/Q	0.118
	net (fo=1, routed)	0.414
1.936	Output[0][0]_OBUF[18]	
C22	OBUF (Prop_obuf_I_O)	1.365
3.301 r	Output[0][0][18]_INST_0/O	
	net (fo=0)	0.000
3.301	Output[0][0][18]	
C22		
r	Output[0][0][18] (OUT)	

```

-----
Slack:                                inf
Source:                               Output_Reg_reg[0][0][9]/C
                                     (rising edge-triggered cell FDRE
clocked by Clock {rise@0.000ns fall@5.000ns period=10.000ns})
Destination:                          Output[0][0][9]
                                     (output port)

Path Group:                           (none)
Path Type:                            Min at Fast Process Corner
Data Path Delay:                       1.918ns  (logic 1.438ns (74.969%)  route
0.480ns (25.031%))
Logic Levels:                          1  (OBUF=1)
Clock Uncertainty:                     0.025ns  ((TSJ^2 + TIJ^2)^1/2 + DJ) / 2 +
PE

```

```

Total System Jitter      (TSJ):      0.050ns
Total Input Jitter       (TIJ):      0.000ns
Discrete Jitter          (DJ):       0.000ns
Phase Error              (PE):       0.000ns

```

Location	Delay type	Incr(ns)
Path(ns)	Netlist Resource(s)	

```

-----
                                     (clock Clock rise edge)      0.000
0.000 r
    D23                                     0.000
0.000 r  Clock (IN)
                                     net (fo=0)                0.000
0.000    Clock
    D23                                     IBUF (Prop_ibuf_I_O)    0.155
0.155 r  Clock_IBUF_inst/O
                                     net (fo=1, routed)        0.676
0.831    Clock_IBUF
    BUFGCTRL_X0Y0                         BUFG (Prop_bufg_I_O)    0.026
0.857 r  Clock_IBUF_BUFG_inst/O

```

```

net (fo=230, routed) 0.545
1.402 Clock_IBUF_BUFG
      SLICE_X9Y82      FDRE
r Output_Reg_reg[0][0][9]/C
-----
-----
      SLICE_X9Y82      FDRE (Prop_fdre_C_Q) 0.100
1.502 r Output_Reg_reg[0][0][9]/Q
      net (fo=1, routed) 0.480
1.982 Output[0][0]_OBUF[9]
      E23      OBUF (Prop_obuf_I_O) 1.338
3.320 r Output[0][0][9]_INST_0/O
      net (fo=0) 0.000
3.320 Output[0][0][9]
      E23
r Output[0][0][9] (OUT)
-----
-----

Slack: inf
Source: Output_Reg_reg[0][0][19]/C
        (rising edge-triggered cell FDRE
clocked by Clock {rise@0.000ns fall@5.000ns period=10.000ns})
Destination: Output[0][0][19]
              (output port)
Path Group: (none)
Path Type: Min at Fast Process Corner
Data Path Delay: 1.931ns (logic 1.481ns (76.690%) route
0.450ns (23.310%))
Logic Levels: 1 (OBUF=1)
Clock Uncertainty: 0.025ns ((TSJ^2 + TIJ^2)^1/2 + DJ) / 2 +
PE
Total System Jitter (TSJ): 0.050ns
Total Input Jitter (TIJ): 0.000ns
Discrete Jitter (DJ): 0.000ns

```

Phase Error (PE) : 0.000ns

Location	Delay type	Incr(ns)
Path(ns) Netlist Resource(s)		

	(clock Clock rise edge)	0.000
0.000 r		
D23		0.000
0.000 r	Clock (IN)	
	net (fo=0)	0.000
0.000	Clock	
D23	IBUF (Prop_ibuf_I_O)	0.155
0.155 r	Clock_IBUF_inst/O	
	net (fo=1, routed)	0.676
0.831	Clock_IBUF	
BUFGCTRL_X0Y0	BUFG (Prop_bufg_I_O)	0.026
0.857 r	Clock_IBUF_BUFG_inst/O	
	net (fo=230, routed)	0.547
1.404	Clock_IBUF_BUFG	
SLICE_X8Y84	FDRE	
r	Output_Reg_reg[0][0][19]/C	

SLICE_X8Y84	FDRE (Prop_fdre_C_Q)	0.118
1.522 r	Output_Reg_reg[0][0][19]/Q	
	net (fo=1, routed)	0.450
1.972	Output[0][0]_OBUF[19]	
D21	OBUF (Prop_obuf_I_O)	1.363
3.335 r	Output[0][0][19]_INST_0/O	
	net (fo=0)	0.000
3.335	Output[0][0][19]	
D21		
r	Output[0][0][19] (OUT)	

```

-----
Slack:                                inf
Source:                              Output_Reg_reg[0][0][10]/C
                                     (rising edge-triggered cell FDRE
clocked by Clock {rise@0.000ns fall@5.000ns period=10.000ns})
Destination:                         Output[0][0][10]
                                     (output port)

Path Group:                          (none)
Path Type:                           Min at Fast Process Corner
Data Path Delay:                     1.936ns  (logic 1.452ns (74.988%)  route
0.484ns (25.012%))
Logic Levels:                        1  (OBUF=1)
Clock Uncertainty:                   0.025ns  ((TSJ^2 + TIJ^2)^1/2 + DJ) / 2 +
PE

```

```

Total System Jitter      (TSJ):      0.050ns
Total Input Jitter       (TIJ):      0.000ns
Discrete Jitter          (DJ):       0.000ns
Phase Error              (PE):       0.000ns

```

Location	Delay type	Incr(ns)
Path(ns)	Netlist Resource(s)	

```

-----
                                (clock Clock rise edge)      0.000
0.000 r
    D23                                0.000
0.000 r  Clock (IN)
                                net (fo=0)      0.000
0.000    Clock
    D23                                IBUF (Prop_ibuf_I_O)      0.155
0.155 r  Clock_IBUF_inst/O
                                net (fo=1, routed)      0.676
0.831    Clock_IBUF
    BUFGCTRL_X0Y0                    BUFG (Prop_bufg_I_O)      0.026

```


0.857	r	Clock_IBUF_BUFG_inst/O			
			net (fo=230, routed)		0.545
1.402		Clock_IBUF_BUFG			
		SLICE_X10Y82	FDRE		
	r	Output_Reg_reg[0][0][10]/C			

		SLICE_X10Y82	FDRE (Prop_fdre_C_Q)		0.118
1.520	r	Output_Reg_reg[0][0][10]/Q			
			net (fo=1, routed)		0.484
2.005		Output[0][0]_OBUF[10]			
		F22	OBUF (Prop_obuf_I_O)		1.334
3.338	r	Output[0][0][10]_INST_0/O			
			net (fo=0)		0.000
3.338		Output[0][0][10]			
		F22			
	r	Output[0][0][10] (OUT)			

Path Group: (none)					
From Clock:					
To Clock: Clock					
Max Delay 332 Endpoints					
Min Delay 332 Endpoints					

Max Delay Paths

Slack: inf
Source: Reset
(input port)
Destination: Output_Reg_reg[0][0][19]/CE
(rising edge-triggered cell FDRE
clocked by Clock {rise@0.000ns fall@5.000ns period=10.000ns})
Path Group: (none)
Path Type: Setup (Max at Slow Process Corner)
Data Path Delay: 3.706ns (logic 0.945ns (25.485%) route
2.762ns (74.515%))
Logic Levels: 2 (IBUF=1 LUT4=1)
Clock Path Skew: 3.890ns (DCD - SCD + CPR)
Destination Clock Delay (DCD): 3.890ns
Source Clock Delay (SCD): 0.000ns
Clock Pessimism Removal (CPR): 0.000ns
Clock Uncertainty: 0.025ns ((TSJ^2 + TIJ^2)^1/2 + DJ) / 2 +
PE
Total System Jitter (TSJ): 0.050ns
Total Input Jitter (TIJ): 0.000ns
Discrete Jitter (DJ): 0.000ns
Phase Error (PE): 0.000ns

Location		Delay type	Incr(ns)
Path(ns)	Netlist Resource(s)		

J26			0.000
0.000 f	Reset (IN)		
		net (fo=0)	0.000
0.000	Reset		
J26		IBUF (Prop_ibuf_I_O)	0.881
0.881 f	Reset_IBUF_inst/O		

	net (fo=42, routed)	1.907
2.788	Reset_IBUF	
	SLICE_X12Y77 LUT4 (Prop_lut4_I0_O)	0.064
2.852	r Output_Reg[0][0][31]_i_1/O	
	net (fo=128, routed)	0.854
3.706	Output_Reg[0][0][31]_i_1_n_0	
	SLICE_X8Y84 FDRE	
	r Output_Reg_reg[0][0][19]/CE	

	(clock Clock rise edge)	0.000
0.000	r	
	D23	0.000
0.000	r Clock (IN)	
	net (fo=0)	0.000
0.000	Clock	
	D23 IBUF (Prop_ibuf_I_O)	0.815
0.815	r Clock_IBUF_inst/O	
	net (fo=1, routed)	1.679
2.494	Clock_IBUF	
	BUFGCTRL_X0Y0 BUFG (Prop_bufg_I_O)	0.113
2.607	r Clock_IBUF_BUFG_inst/O	
	net (fo=230, routed)	1.283
3.890	Clock_IBUF_BUFG	
	SLICE_X8Y84 FDRE	
	r Output_Reg_reg[0][0][19]/C	
Slack:	inf	
Source:	Reset	
	(input port)	
Destination:	Output_Reg_reg[1][1][18]/CE	
	(rising edge-triggered cell FDRE	
clocked by Clock {rise@0.000ns fall@5.000ns period=10.000ns})		
Path Group:	(none)	
Path Type:	Setup (Max at Slow Process Corner)	

```

Data Path Delay:          3.668ns (logic 0.945ns (25.751%) route
2.723ns (74.249%))
Logic Levels:            2  (IBUF=1 LUT4=1)
Clock Path Skew:         3.886ns (DCD - SCD + CPR)
  Destination Clock Delay (DCD):    3.886ns
  Source Clock Delay      (SCD):    0.000ns
  Clock Pessimism Removal (CPR):    0.000ns
Clock Uncertainty:       0.025ns ((TSJ^2 + TIJ^2)^1/2 + DJ) / 2 +
PE
  Total System Jitter      (TSJ):    0.050ns
  Total Input Jitter       (TIJ):    0.000ns
  Discrete Jitter          (DJ):    0.000ns
  Phase Error              (PE):    0.000ns

```

Location	Delay type	Incr(ns)
Path(ns) Netlist Resource(s)		

J26		0.000
0.000 f Reset (IN)		
	net (fo=0)	0.000
0.000 Reset		
J26	IBUF (Prop_ibuf_I_O)	0.881
0.881 f Reset_IBUF_inst/O		
	net (fo=42, routed)	1.907
2.788 Reset_IBUF		
SLICE_X12Y77	LUT4 (Prop_lut4_I0_O)	0.064
2.852 r Output_Reg[0][0][31]_i_1/O		
	net (fo=128, routed)	0.816
3.668 Output_Reg[0][0][31]_i_1_n_0		
SLICE_X9Y80	FDRE	
r Output_Reg_reg[1][1][18]/CE		

```

                                (clock Clock rise edge)                0.000
0.000 r
      D23                                                                0.000
0.000 r  Clock (IN)
                                net (fo=0)                            0.000
0.000      Clock
      D23                      IBUF (Prop_ibuf_I_O)                    0.815
0.815 r  Clock_IBUF_inst/O
                                net (fo=1, routed)                    1.679
2.494      Clock_IBUF
      BUFGCTRL_X0Y0            BUFG (Prop_bufg_I_O)                    0.113
2.607 r  Clock_IBUF_BUFG_inst/O
                                net (fo=230, routed)                  1.279
3.886      Clock_IBUF_BUFG
      SLICE_X9Y80              FDRE
r  Output_Reg_reg[1][1][18]/C

Slack:                          inf
Source:                          Reset
                                (input port)
Destination:                     Output_Reg_reg[1][1][29]/CE
                                (rising edge-triggered cell FDRE
clocked by Clock {rise@0.000ns fall@5.000ns period=10.000ns})
Path Group:                      (none)
Path Type:                       Setup (Max at Slow Process Corner)
Data Path Delay:                  3.668ns  (logic 0.945ns (25.751%)  route
2.723ns (74.249%))
Logic Levels:                     2  (IBUF=1 LUT4=1)
Clock Path Skew:                  3.886ns (DCD - SCD + CPR)
  Destination Clock Delay (DCD):    3.886ns
  Source Clock Delay (SCD):          0.000ns
  Clock Pessimism Removal (CPR):    0.000ns
Clock Uncertainty:                0.025ns  ((TSJ^2 + TIJ^2)^1/2 + DJ) / 2 +
PE
Total System Jitter (TSJ):         0.050ns
Total Input Jitter (TIJ):          0.000ns
Discrete Jitter (DJ):              0.000ns

```

Phase Error (PE): 0.000ns

Location	Delay type	Incr(ns)
Path(ns) Netlist Resource(s)		

J26		0.000
0.000 f Reset (IN)		
	net (fo=0)	0.000
0.000 Reset		
J26	IBUF (Prop_ibuf_I_O)	0.881
0.881 f Reset_IBUF_inst/O		
	net (fo=42, routed)	1.907
2.788 Reset_IBUF		
SLICE_X12Y77	LUT4 (Prop_lut4_I0_O)	0.064
2.852 r Output_Reg[0][0][31]_i_1/O		
	net (fo=128, routed)	0.816
3.668 Output_Reg[0][0][31]_i_1_n_0		
SLICE_X9Y80	FDRE	
r Output_Reg_reg[1][1][29]/CE		

	(clock Clock rise edge)	0.000
0.000 r		
D23		0.000
0.000 r Clock (IN)		
	net (fo=0)	0.000
0.000 Clock		
D23	IBUF (Prop_ibuf_I_O)	0.815
0.815 r Clock_IBUF_inst/O		
	net (fo=1, routed)	1.679
2.494 Clock_IBUF		
BUFGCTRL_X0Y0	BUFG (Prop_bufg_I_O)	0.113
2.607 r Clock_IBUF_BUFG_inst/O		

```

net (fo=230, routed) 1.279
3.886 Clock_IBUF_BUFG
SLICE_X9Y80 FDRE
r Output_Reg_reg[1][1][29]/C

Slack: inf
Source: Reset
(input port)
Destination: Output_Reg_reg[1][1][7]/CE
(rising edge-triggered cell FDRE
clocked by Clock {rise@0.000ns fall@5.000ns period=10.000ns})
Path Group: (none)
Path Type: Setup (Max at Slow Process Corner)
Data Path Delay: 3.668ns (logic 0.945ns (25.751%) route
2.723ns (74.249%))
Logic Levels: 2 (IBUF=1 LUT4=1)
Clock Path Skew: 3.886ns (DCD - SCD + CPR)
Destination Clock Delay (DCD): 3.886ns
Source Clock Delay (SCD): 0.000ns
Clock Pessimism Removal (CPR): 0.000ns
Clock Uncertainty: 0.025ns ((TSJ^2 + TIJ^2)^1/2 + DJ) / 2 +
PE
Total System Jitter (TSJ): 0.050ns
Total Input Jitter (TIJ): 0.000ns
Discrete Jitter (DJ): 0.000ns
Phase Error (PE): 0.000ns

```

Location	Delay type	Incr(ns)
Path(ns)	Netlist Resource(s)	

J26		0.000
0.000 f Reset (IN)		
	net (fo=0)	0.000
0.000 Reset		
J26	IBUF (Prop_ibuf_I_O)	0.881

```

0.881 f  Reset_IBUF_inst/O
                                net (fo=42, routed)                1.907
2.788      Reset_IBUF
      SLICE_X12Y77              LUT4 (Prop_lut4_I0_O)              0.064
2.852 r  Output_Reg[0][0][31]_i_1/O
                                net (fo=128, routed)              0.816
3.668      Output_Reg[0][0][31]_i_1_n_0
      SLICE_X9Y80              FDRE
r  Output_Reg_reg[1][1][7]/CE
-----
-----
                                (clock Clock rise edge)          0.000
0.000 r
      D23                      0.000
0.000 r  Clock (IN)
                                net (fo=0)                        0.000
0.000      Clock
      D23                      IBUF (Prop_ibuf_I_O)              0.815
0.815 r  Clock_IBUF_inst/O
                                net (fo=1, routed)                1.679
2.494      Clock_IBUF
      BUFGCTRL_X0Y0            BUFG (Prop_bufg_I_O)              0.113
2.607 r  Clock_IBUF_BUFG_inst/O
                                net (fo=230, routed)              1.279
3.886      Clock_IBUF_BUFG
      SLICE_X9Y80              FDRE
r  Output_Reg_reg[1][1][7]/C

Slack:                          inf
Source:                          Reset
                                (input port)
Destination:                     Output_Reg_reg[1][1][8]/CE
                                (rising edge-triggered cell FDRE
clocked by Clock {rise@0.000ns fall@5.000ns period=10.000ns})
Path Group:                      (none)

```



```

Path Type:                               Setup (Max at Slow Process Corner)
Data Path Delay:                         3.668ns  (logic 0.945ns (25.751%)  route
2.723ns (74.249%))
Logic Levels:                           2  (IBUF=1 LUT4=1)
Clock Path Skew:                         3.886ns (DCD - SCD + CPR)
  Destination Clock Delay (DCD):         3.886ns
  Source Clock Delay (SCD):               0.000ns
  Clock Pessimism Removal (CPR):         0.000ns
Clock Uncertainty:                       0.025ns  ((TSJ^2 + TIJ^2)^1/2 + DJ) / 2 +
PE
  Total System Jitter (TSJ):             0.050ns
  Total Input Jitter (TIJ):              0.000ns
  Discrete Jitter (DJ):                  0.000ns
  Phase Error (PE):                      0.000ns

```

Location	Delay type	Incr(ns)
Path(ns) Netlist Resource(s)		

J26		0.000
0.000 f Reset (IN)		
	net (fo=0)	0.000
0.000 Reset		
J26	IBUF (Prop_ibuf_I_O)	0.881
0.881 f Reset_IBUF_inst/O		
	net (fo=42, routed)	1.907
2.788 Reset_IBUF		
SLICE_X12Y77	LUT4 (Prop_lut4_I0_O)	0.064
2.852 r Output_Reg[0][0][31]_i_1/O		
	net (fo=128, routed)	0.816
3.668 Output_Reg[0][0][31]_i_1_n_0		
SLICE_X9Y80	FDRE	
r Output_Reg_reg[1][1][8]/CE		


```

                                (clock Clock rise edge)          0.000
0.000 r
    D23                                0.000
0.000 r  Clock (IN)
                                net (fo=0)                      0.000
0.000    Clock
    D23                                IBUF (Prop_ibuf_I_O)      0.815
0.815 r  Clock_IBUF_inst/O
                                net (fo=1, routed)             1.679
2.494    Clock_IBUF
    BUFGCTRL_X0Y0                    BUFG (Prop_bufg_I_O)      0.113
2.607 r  Clock_IBUF_BUFG_inst/O
                                net (fo=230, routed)           1.279
3.886    Clock_IBUF_BUFG
    SLICE_X9Y80                      FDRE
r  Output_Reg_reg[1][1][8]/C

Slack:                            inf
Source:                            Reset
                                (input port)
Destination:                       Output_Reg_reg[1][1][9]/CE
                                (rising edge-triggered cell FDRE
clocked by Clock {rise@0.000ns fall@5.000ns period=10.000ns})
Path Group:                        (none)
Path Type:                         Setup (Max at Slow Process Corner)
Data Path Delay:                   3.668ns  (logic 0.945ns (25.751%)  route
2.723ns (74.249%))
Logic Levels:                      2  (IBUF=1 LUT4=1)
Clock Path Skew:                   3.886ns (DCD - SCD + CPR)
    Destination Clock Delay (DCD):    3.886ns
    Source Clock Delay (SCD):         0.000ns
    Clock Pessimism Removal (CPR):    0.000ns
Clock Uncertainty:                 0.025ns  ((TSJ^2 + TIJ^2)^1/2 + DJ) / 2 +
PE
    Total System Jitter (TSJ):        0.050ns
    Total Input Jitter (TIJ):         0.000ns

```

Discrete Jitter		(DJ):	0.000ns
Phase Error		(PE):	0.000ns
Location		Delay type	Incr(ns)
Path(ns)	Netlist Resource(s)		

J26			0.000
0.000 f	Reset (IN)		
	net (fo=0)		0.000
0.000	Reset		
J26		IBUF (Prop_ibuf_I_O)	0.881
0.881 f	Reset_IBUF_inst/O		
	net (fo=42, routed)		1.907
2.788	Reset_IBUF		
SLICE_X12Y77		LUT4 (Prop_lut4_I0_O)	0.064
2.852 r	Output_Reg[0][0][31]_i_1/O		
	net (fo=128, routed)		0.816
3.668	Output_Reg[0][0][31]_i_1_n_0		
SLICE_X9Y80		FDRE	
r	Output_Reg_reg[1][1][9]/CE		

	(clock Clock rise edge)		0.000
0.000 r			
D23			0.000
0.000 r	Clock (IN)		
	net (fo=0)		0.000
0.000	Clock		
D23		IBUF (Prop_ibuf_I_O)	0.815
0.815 r	Clock_IBUF_inst/O		
	net (fo=1, routed)		1.679
2.494	Clock_IBUF		
BUFGCTRL_X0Y0		BUFG (Prop_bufg_I_O)	0.113

```

2.607 r Clock_IBUF_BUFG_inst/O
                                net (fo=230, routed)                1.279

3.886 Clock_IBUF_BUFG
  SLICE_X9Y80 FDRE
r Output_Reg_reg[1][1][9]/C

Slack: inf
Source: A_Write_Data[0]
        (input port)
Destination: A_Reg_reg[1][1][0]/D
              (rising edge-triggered cell FDRE
clocked by Clock {rise@0.000ns fall@5.000ns period=10.000ns})
Path Group: (none)
Path Type: Setup (Max at Slow Process Corner)
Data Path Delay: 3.662ns (logic 0.802ns (21.890%) route
2.860ns (78.110%))
Logic Levels: 1 (IBUF=1)
Clock Path Skew: 3.882ns (DCD - SCD + CPR)
  Destination Clock Delay (DCD): 3.882ns
  Source Clock Delay (SCD): 0.000ns
  Clock Pessimism Removal (CPR): 0.000ns
Clock Uncertainty: 0.025ns ((TSJ^2 + TIJ^2)^1/2 + DJ) / 2 +
PE
  Total System Jitter (TSJ): 0.050ns
  Total Input Jitter (TIJ): 0.000ns
  Discrete Jitter (DJ): 0.000ns
  Phase Error (PE): 0.000ns

Location Delay type Incr(ns)
Path(ns) Netlist Resource(s)
-----
-----
U16 0.000 0.000
0.000 r A_Write_Data[0] (IN)
                                net (fo=0) 0.000
0.000 A_Write_Data[0]

```

```

U16                                IBUF (Prop_ibuf_I_O)                0.802
0.802 r  A_Write_Data_IBUF[0]_inst/O
                                net (fo=4, routed)                2.860
3.662    A_Write_Data_IBUF[0]
    SLICE_X12Y73                FDRE
r  A_Reg_reg[1][1][0]/D

-----

                                (clock Clock rise edge)            0.000
0.000 r
    D23                          0.000
0.000 r  Clock (IN)
                                net (fo=0)                        0.000
0.000    Clock
    D23                          IBUF (Prop_ibuf_I_O)            0.815
0.815 r  Clock_IBUF_inst/O
                                net (fo=1, routed)                1.679
2.494    Clock_IBUF
    BUFGCTRL_X0Y0                BUFG (Prop_bufg_I_O)            0.113
2.607 r  Clock_IBUF_BUFG_inst/O
                                net (fo=230, routed)              1.275
3.882    Clock_IBUF_BUFG
    SLICE_X12Y73                FDRE
r  A_Reg_reg[1][1][0]/C

Slack:                            inf
Source:                            Reset
                                (input port)
Destination:                      Output_Reg_reg[0][1][28]/CE
                                (rising edge-triggered cell FDRE
clocked by Clock {rise@0.000ns fall@5.000ns period=10.000ns})
Path Group:                        (none)
Path Type:                        Setup (Max at Slow Process Corner)
Data Path Delay:                   3.637ns (logic 0.945ns (25.966%) route
2.693ns (74.034%))

```

```

Logic Levels:                2    (IBUF=1 LUT4=1)
Clock Path Skew:              3.889ns (DCD - SCD + CPR)
    Destination Clock Delay (DCD):    3.889ns
    Source Clock Delay      (SCD):    0.000ns
    Clock Pessimism Removal (CPR):    0.000ns
Clock Uncertainty:            0.025ns  ((TSJ^2 + TIJ^2)^1/2 + DJ) / 2 +
PE
    Total System Jitter      (TSJ):    0.050ns
    Total Input Jitter       (TIJ):    0.000ns
    Discrete Jitter          (DJ):    0.000ns
    Phase Error              (PE):    0.000ns

Location                      Delay type                      Incr(ns)
Path(ns)    Netlist Resource(s)

-----
-----
    J26                                              0.000
0.000 f  Reset (IN)
                      net (fo=0)                      0.000
0.000    Reset
    J26                      IBUF (Prop_ibuf_I_O)      0.881
0.881 f  Reset_IBUF_inst/O
                      net (fo=42, routed)             1.907
2.788    Reset_IBUF
    SLICE_X12Y77          LUT4 (Prop_lut4_I0_O)        0.064
2.852 r  Output_Reg[0][0][31]_i_1/O
                      net (fo=128, routed)             0.786
3.637    Output_Reg[0][0][31]_i_1_n_0
    SLICE_X8Y82            FDRE
r  Output_Reg_reg[0][1][28]/CE

-----
-----
                      (clock Clock rise edge)        0.000
0.000 r

```

```

D23 0.000 r Clock (IN) 0.000
net (fo=0) 0.000
0.000 Clock
D23 IBUF (Prop_ibuf_I_O) 0.815
0.815 r Clock_IBUF_inst/O
net (fo=1, routed) 1.679
2.494 Clock_IBUF
BUFGCTRL_X0Y0 BUFG (Prop_bufg_I_O) 0.113
2.607 r Clock_IBUF_BUFG_inst/O
net (fo=230, routed) 1.282
3.889 Clock_IBUF_BUFG
SLICE_X8Y82 FDRE
r Output_Reg_reg[0][1][28]/C

Slack: inf
Source: Reset
(input port)
Destination: Output_Reg_reg[0][1][31]/CE
(rising edge-triggered cell FDRE
clocked by Clock {rise@0.000ns fall@5.000ns period=10.000ns})
Path Group: (none)
Path Type: Setup (Max at Slow Process Corner)
Data Path Delay: 3.637ns (logic 0.945ns (25.966%) route
2.693ns (74.034%))
Logic Levels: 2 (IBUF=1 LUT4=1)
Clock Path Skew: 3.889ns (DCD - SCD + CPR)
Destination Clock Delay (DCD): 3.889ns
Source Clock Delay (SCD): 0.000ns
Clock Pessimism Removal (CPR): 0.000ns
Clock Uncertainty: 0.025ns ((TSJ^2 + TIJ^2)^1/2 + DJ) / 2 +
PE
Total System Jitter (TSJ): 0.050ns
Total Input Jitter (TIJ): 0.000ns
Discrete Jitter (DJ): 0.000ns
Phase Error (PE): 0.000ns

```

Location		Delay type		Incr(ns)
Path(ns)	Netlist	Resource(s)		

	J26			0.000
0.000 f	Reset	(IN)		
		net	(fo=0)	0.000
0.000	Reset			
	J26	IBUF	(Prop_ibuf_I_O)	0.881
0.881 f	Reset_IBUF_inst/O			
		net	(fo=42, routed)	1.907
2.788	Reset_IBUF			
	SLICE_X12Y77	LUT4	(Prop_lut4_I0_O)	0.064
2.852 r	Output_Reg[0][0][31]_i_1/O			
		net	(fo=128, routed)	0.786
3.637	Output_Reg[0][0][31]_i_1_n_0			
	SLICE_X8Y82	FDRE		
r	Output_Reg_reg[0][1][31]/CE			

		(clock	Clock rise edge)	0.000
0.000 r				
	D23			0.000
0.000 r	Clock	(IN)		
		net	(fo=0)	0.000
0.000	Clock			
	D23	IBUF	(Prop_ibuf_I_O)	0.815
0.815 r	Clock_IBUF_inst/O			
		net	(fo=1, routed)	1.679
2.494	Clock_IBUF			
	BUFGCTRL_X0Y0	BUFG	(Prop_bufg_I_O)	0.113
2.607 r	Clock_IBUF_BUFG_inst/O			
		net	(fo=230, routed)	1.282
3.889	Clock_IBUF_BUFG			


```

SLICE_X8Y82          FDRE
r  Output_Reg_reg[0][1][31]/C

Slack:               inf
Source:              Reset
                    (input port)
Destination:         Output_Reg_reg[0][1][8]/CE
                    (rising edge-triggered cell FDRE
clocked by Clock {rise@0.000ns fall@5.000ns period=10.000ns})
Path Group:          (none)
Path Type:           Setup (Max at Slow Process Corner)
Data Path Delay:      3.637ns  (logic 0.945ns (25.966%)  route
2.693ns (74.034%))
Logic Levels:        2  (IBUF=1 LUT4=1)
Clock Path Skew:      3.889ns (DCD - SCD + CPR)
  Destination Clock Delay (DCD):    3.889ns
  Source Clock Delay      (SCD):    0.000ns
  Clock Pessimism Removal (CPR):    0.000ns
Clock Uncertainty:    0.025ns  ((TSJ^2 + TIJ^2)^1/2 + DJ) / 2 +
PE
  Total System Jitter      (TSJ):    0.050ns
  Total Input Jitter       (TIJ):    0.000ns
  Discrete Jitter          (DJ):    0.000ns
  Phase Error              (PE):    0.000ns

```

Location	Delay type	Incr(ns)
Path(ns)	Netlist Resource(s)	

J26		0.000
0.000 f Reset (IN)		
	net (fo=0)	0.000
0.000 Reset		
J26	IBUF (Prop_ibuf_I_O)	0.881
0.881 f Reset_IBUF_inst/O		
	net (fo=42, routed)	1.907

2.788	Reset_IBUF		
	SLICE_X12Y77	LUT4 (Prop_lut4_I0_O)	0.064
2.852	r Output_Reg[0][0][31]_i_1/O		
		net (fo=128, routed)	0.786
3.637	Output_Reg[0][0][31]_i_1_n_0		
	SLICE_X8Y82	FDRE	
	r Output_Reg_reg[0][1][8]/CE		

		(clock Clock rise edge)	0.000
0.000	r		
	D23		0.000
0.000	r Clock (IN)		
		net (fo=0)	0.000
0.000	Clock		
	D23	IBUF (Prop_ibuf_I_O)	0.815
0.815	r Clock_IBUF_inst/O		
		net (fo=1, routed)	1.679
2.494	Clock_IBUF		
	BUFGCTRL_X0Y0	BUFG (Prop_bufg_I_O)	0.113
2.607	r Clock_IBUF_BUFG_inst/O		
		net (fo=230, routed)	1.282
3.889	Clock_IBUF_BUFG		
	SLICE_X8Y82	FDRE	
	r Output_Reg_reg[0][1][8]/C		

Min Delay Paths

```

Slack:                inf
Source:               B_Write_Data[6]

```

```

        (input port)
Destination:          B_Reg_reg[1][0][6]/D
                      (rising edge-triggered cell FDRE
clocked by Clock {rise@0.000ns fall@5.000ns period=10.000ns})
Path Group:          (none)
Path Type:           Hold (Min at Fast Process Corner)
Data Path Delay:      0.818ns (logic 0.132ns (16.122%) route
0.686ns (83.878%))
Logic Levels:        1 (IBUF=1)
Clock Path Skew:      1.908ns (DCD - SCD - CPR)
  Destination Clock Delay (DCD):    1.908ns
  Source Clock Delay (SCD):         0.000ns
  Clock Pessimism Removal (CPR):    -0.000ns

```

Location	Delay type	Incr(ns)
Path(ns)	Netlist Resource(s)	

```

-----
-----
J24 0.000 r B_Write_Data[6] (IN) 0.000
net (fo=0) 0.000
B_Write_Data[6]
J24 0.132 r B_Write_Data_IBUF[6]_inst/O 0.132
net (fo=4, routed) 0.686
B_Write_Data_IBUF[6]
SLICE_X13Y75 FDRE
r B_Reg_reg[1][0][6]/D
-----
-----
(clock Clock rise edge) 0.000
0.000 r
D23 0.000
0.000 r Clock (IN)

```

```

                                net (fo=0)                                0.000
0.000      Clock
      D23                        IBUF (Prop_ibuf_I_O)                    0.404
0.404 r    Clock_IBUF_inst/O
                                net (fo=1, routed)                    0.739
1.143      Clock_IBUF
      BUFGCTRL_X0Y0            BUFG (Prop_bufg_I_O)                    0.030
1.173 r    Clock_IBUF_BUFG_inst/O
                                net (fo=230, routed)                  0.735
1.908      Clock_IBUF_BUFG
      SLICE_X13Y75             FDRE
r    B_Reg_reg[1][0][6]/C

Slack:                                inf
Source:                                Reset
                                (input port)
Destination:                         FSM_sequential_State_reg[0]/D
                                (rising edge-triggered cell FDRE
clocked by Clock {rise@0.000ns fall@5.000ns period=10.000ns})
Path Group:                          (none)
Path Type:                           Hold (Min at Fast Process Corner)
Data Path Delay:                      0.843ns (logic 0.172ns (20.422%) route
0.670ns (79.578%))
Logic Levels:                        2 (IBUF=1 LUT3=1)
Clock Path Skew:                     1.912ns (DCD - SCD - CPR)
  Destination Clock Delay (DCD):      1.912ns
  Source Clock Delay (SCD):           0.000ns
  Clock Pessimism Removal (CPR):      -0.000ns

Location          Delay type          Incr(ns)
Path(ns)          Netlist Resource(s)
-----
-----
      J26                                0.000
0.000 f    Reset (IN)
                                net (fo=0)                                0.000

```

0.000	Reset		
	J26	IBUF (Prop_ibuf_I_O)	0.144
0.144	f	Reset_IBUF_inst/O	
		net (fo=42, routed)	0.670
0.815	Reset_IBUF		
	SLICE_X8Y78	LUT3 (Prop_lut3_I2_O)	0.028
0.843	r	FSM_sequential_State[0]_i_1/O	
		net (fo=1, routed)	0.000
0.843	FSM_sequential_State[0]_i_1_n_0		
	SLICE_X8Y78	FDRE	
	r	FSM_sequential_State_reg[0]/D	

		(clock Clock rise edge)	0.000
0.000	r		
	D23		0.000
0.000	r	Clock (IN)	
		net (fo=0)	0.000
0.000	Clock		
	D23	IBUF (Prop_ibuf_I_O)	0.404
0.404	r	Clock_IBUF_inst/O	
		net (fo=1, routed)	0.739
1.143	Clock_IBUF		
	BUFGCTRL_X0Y0	BUFG (Prop_bufg_I_O)	0.030
1.173	r	Clock_IBUF_BUFG_inst/O	
		net (fo=230, routed)	0.739
1.912	Clock_IBUF_BUFG		
	SLICE_X8Y78	FDRE	
	r	FSM_sequential_State_reg[0]/C	
Slack:		inf	
Source:		Reset	
		(input port)	
Destination:		FSM_sequential_State_reg[1]/D	
		(rising edge-triggered cell FDRE	

clocked by Clock {rise@0.000ns fall@5.000ns period=10.000ns})

Path Group: (none)

Path Type: Hold (Min at Fast Process Corner)

Data Path Delay: 0.845ns (logic 0.174ns (20.611%) route 0.670ns (79.389%))

Logic Levels: 2 (IBUF=1 LUT4=1)

Clock Path Skew: 1.912ns (DCD - SCD - CPR)

Destination Clock Delay (DCD): 1.912ns

Source Clock Delay (SCD): 0.000ns

Clock Pessimism Removal (CPR): -0.000ns

Location	Delay type	Incr(ns)
Path(ns)	Netlist Resource(s)	

J26		0.000
0.000 f Reset (IN)		
	net (fo=0)	0.000
0.000 Reset		
J26	IBUF (Prop_ibuf_I_O)	0.144
0.144 f Reset_IBUF_inst/O		
	net (fo=42, routed)	0.670
0.815 Reset_IBUF		
SLICE_X8Y78	LUT4 (Prop_lut4_I3_O)	0.030
0.845 r FSM_sequential_State[1]_i_1/O		
	net (fo=1, routed)	0.000
0.845 FSM_sequential_State[1]_i_1_n_0		
SLICE_X8Y78	FDRE	
r FSM_sequential_State_reg[1]/D		

	(clock Clock rise edge)	0.000
0.000 r		
D23		0.000

		net (fo=0)	0.000
0.000	Reset		
	J26	IBUF (Prop_ibuf_I_O)	0.144
0.144	r Reset_IBUF_inst/O		
		net (fo=42, routed)	0.701
0.846	Reset_IBUF		
	SLICE_X9Y77	FDRE	
	r Cycle_Count_reg[11]/R		

		(clock Clock rise edge)	0.000
0.000	r		
	D23		0.000
0.000	r Clock (IN)		
		net (fo=0)	0.000
0.000	Clock		
	D23	IBUF (Prop_ibuf_I_O)	0.404
0.404	r Clock_IBUF_inst/O		
		net (fo=1, routed)	0.739
1.143	Clock_IBUF		
	BUFGCTRL_X0Y0	BUFG (Prop_bufg_I_O)	0.030
1.173	r Clock_IBUF_BUFG_inst/O		
		net (fo=230, routed)	0.738
1.911	Clock_IBUF_BUFG		
	SLICE_X9Y77	FDRE	
	r Cycle_Count_reg[11]/C		
Slack:		inf	
Source:		Reset	
		(input port)	
Destination:		Cycle_Count_reg[10]/R	
		(rising edge-triggered cell FDRE	
clocked by Clock {rise@0.000ns fall@5.000ns period=10.000ns})			
Path Group:		(none)	
Path Type:		Hold (Min at Fast Process Corner)	

Data Path Delay: 0.860ns (logic 0.144ns (16.753%) route
0.716ns (83.247%))

Logic Levels: 1 (IBUF=1)

Clock Path Skew: 1.911ns (DCD - SCD - CPR)

Destination Clock Delay (DCD): 1.911ns

Source Clock Delay (SCD): 0.000ns

Clock Pessimism Removal (CPR): -0.000ns

Location	Delay type	Incr(ns)
Path(ns) Netlist Resource(s)		

J26		0.000
0.000 r Reset (IN)		
	net (fo=0)	0.000
0.000 Reset		
J26	IBUF (Prop_ibuf_I_O)	0.144
0.144 r Reset_IBUF_inst/O		
	net (fo=42, routed)	0.716
0.860 Reset_IBUF		
SLICE_X11Y77	FDRE	
r Cycle_Count_reg[10]/R		

	(clock Clock rise edge)	0.000
0.000 r		
D23		0.000
0.000 r Clock (IN)		
	net (fo=0)	0.000
0.000 Clock		
D23	IBUF (Prop_ibuf_I_O)	0.404
0.404 r Clock_IBUF_inst/O		
	net (fo=1, routed)	0.739
1.143 Clock_IBUF		

```

    BUFGCTRL_X0Y0          BUFG (Prop_bufg_I_O)          0.030
1.173 r  Clock_IBUF_BUFG_inst/O
                                net (fo=230, routed)      0.738
1.911    Clock_IBUF_BUFG
    SLICE_X11Y77          FDRE
r  Cycle_Count_reg[10]/C

Slack:                      inf
Source:                      Reset
                                (input port)
Destination:                Cycle_Count_reg[12]/R
                                (rising edge-triggered cell FDRE
clocked by Clock {rise@0.000ns fall@5.000ns period=10.000ns})
Path Group:                  (none)
Path Type:                   Hold (Min at Fast Process Corner)
Data Path Delay:             0.860ns  (logic 0.144ns (16.753%)  route
0.716ns (83.247%))
Logic Levels:                1  (IBUF=1)
Clock Path Skew:             1.911ns (DCD - SCD - CPR)
    Destination Clock Delay (DCD):      1.911ns
    Source Clock Delay      (SCD):      0.000ns
    Clock Pessimism Removal (CPR):      -0.000ns

```

Location	Delay type	Incr(ns)
Path(ns) Netlist Resource(s)		

J26		0.000
0.000 r Reset (IN)		
	net (fo=0)	0.000
0.000 Reset		
J26	IBUF (Prop_ibuf_I_O)	0.144
0.144 r Reset_IBUF_inst/O		
	net (fo=42, routed)	0.716
0.860 Reset_IBUF		
SLICE_X11Y77	FDRE	

```

r  Cycle_Count_reg[12]/R
-----
-----

                                (clock Clock rise edge)                0.000
0.000 r
    D23                                                                0.000
0.000 r  Clock (IN)
                                net (fo=0)                            0.000
0.000    Clock
    D23                                IBUF (Prop_ibuf_I_O)            0.404
0.404 r  Clock_IBUF_inst/O
                                net (fo=1, routed)                    0.739
1.143    Clock_IBUF
    BUFGCTRL_X0Y0                BUFG (Prop_bufg_I_O)                0.030
1.173 r  Clock_IBUF_BUFG_inst/O
                                net (fo=230, routed)                  0.738
1.911    Clock_IBUF_BUFG
    SLICE_X11Y77                FDRE
r  Cycle_Count_reg[12]/C

Slack:                        inf
Source:                        Reset
                                (input port)
Destination:                  Cycle_Count_reg[9]/R
                                (rising edge-triggered cell FDRE
clocked by Clock {rise@0.000ns fall@5.000ns period=10.000ns})
Path Group:                    (none)
Path Type:                     Hold (Min at Fast Process Corner)
Data Path Delay:               0.860ns  (logic 0.144ns (16.753%)  route
0.716ns (83.247%))
Logic Levels:                  1  (IBUF=1)
Clock Path Skew:               1.911ns (DCD - SCD - CPR)
    Destination Clock Delay (DCD):    1.911ns
    Source Clock Delay      (SCD):     0.000ns
    Clock Pessimism Removal (CPR):    -0.000ns

```

Location	Delay type	Incr(ns)
Path(ns)	Netlist Resource(s)	

J26		0.000
0.000 r Reset (IN)		
	net (fo=0)	0.000
0.000 Reset		
J26	IBUF (Prop_ibuf_I_O)	0.144
0.144 r Reset_IBUF_inst/O		
	net (fo=42, routed)	0.716
0.860 Reset_IBUF		
SLICE_X11Y77	FDRE	
r Cycle_Count_reg[9]/R		

	(clock Clock rise edge)	0.000
0.000 r		
D23		0.000
0.000 r Clock (IN)		
	net (fo=0)	0.000
0.000 Clock		
D23	IBUF (Prop_ibuf_I_O)	0.404
0.404 r Clock_IBUF_inst/O		
	net (fo=1, routed)	0.739
1.143 Clock_IBUF		
BUFGCTRL_X0Y0	BUFG (Prop_bufg_I_O)	0.030
1.173 r Clock_IBUF_BUFG_inst/O		
	net (fo=230, routed)	0.738
1.911 Clock_IBUF_BUFG		
SLICE_X11Y77	FDRE	
r Cycle_Count_reg[9]/C		

```

Slack:                                inf
Source:                               A_Write_Col[0]
                                       (input port)
Destination:                          A_Reg_reg[0][0][5]/CE
                                       (rising edge-triggered cell FDRE
clocked by Clock {rise@0.000ns fall@5.000ns period=10.000ns})
Path Group:                           (none)
Path Type:                            Hold (Min at Fast Process Corner)
Data Path Delay:                      0.885ns  (logic 0.176ns (19.835%)  route
0.710ns (80.165%))
Logic Levels:                         2  (IBUF=1 LUT3=1)
Clock Path Skew:                      1.912ns (DCD - SCD - CPR)
  Destination Clock Delay (DCD):       1.912ns
  Source Clock Delay      (SCD):       0.000ns
  Clock Pessimism Removal (CPR):       -0.000ns

```

Location	Delay type	Incr(ns)
Path(ns)	Netlist Resource(s)	

J21		0.000
0.000 f	A_Write_Col[0] (IN)	
	net (fo=0)	0.000
0.000	A_Write_Col[0]	
J21	IBUF (Prop_ibuf_I_O)	0.146
0.146 f	A_Write_Col_IBUF[0]_inst/O	
	net (fo=4, routed)	0.545
0.690	A_Write_Col_IBUF[0]	
	SLICE_X8Y68 LUT3 (Prop_lut3_I1_O)	0.030
0.720 r	A_Reg[0][0][7]_i_1/O	
	net (fo=8, routed)	0.165
0.885	A_Reg[0][0][0]	
	SLICE_X10Y71 FDRE	
r	A_Reg_reg[0][0][5]/CE	

```

-----
                                (clock Clock rise edge)                0.000
0.000 r
    D23                                                                0.000
0.000 r  Clock (IN)
                                net (fo=0)                            0.000
0.000    Clock
    D23                                IBUF (Prop_ibuf_I_O)            0.404
0.404 r  Clock_IBUF_inst/O
                                net (fo=1, routed)                    0.739
1.143    Clock_IBUF
    BUFGCTRL_X0Y0                BUFG (Prop_bufg_I_O)                0.030
1.173 r  Clock_IBUF_BUFG_inst/O
                                net (fo=230, routed)                  0.739
1.912    Clock_IBUF_BUFG
    SLICE_X10Y71                FDRE
r  A_Reg_reg[0][0][5]/C

Slack:                        inf
Source:                        Reset
                                (input port)
Destination:                  Cycle_Count_reg[0]/R
                                (rising edge-triggered cell FDRE
clocked by Clock {rise@0.000ns fall@5.000ns period=10.000ns})
Path Group:                    (none)
Path Type:                     Hold (Min at Fast Process Corner)
Data Path Delay:                0.890ns  (logic 0.144ns (16.192%)  route
0.746ns (83.808%))
Logic Levels:                  1  (IBUF=1)
Clock Path Skew:                1.912ns (DCD - SCD - CPR)
    Destination Clock Delay (DCD):    1.912ns
    Source Clock Delay      (SCD):     0.000ns
    Clock Pessimism Removal (CPR):    -0.000ns

Location                        Delay type                        Incr(ns)
Path(ns)    Netlist Resource(s)

```


	J26		0.000
0.000	r	Reset (IN)	
		net (fo=0)	0.000
0.000		Reset	
	J26	IBUF (Prop_ibuf_I_O)	0.144
0.144	r	Reset_IBUF_inst/O	
		net (fo=42, routed)	0.746
0.890		Reset_IBUF	
	SLICE_X9Y78	FDRE	
	r	Cycle_Count_reg[0]/R	

		(clock Clock rise edge)	0.000
0.000	r		
	D23		0.000
0.000	r	Clock (IN)	
		net (fo=0)	0.000
0.000		Clock	
	D23	IBUF (Prop_ibuf_I_O)	0.404
0.404	r	Clock_IBUF_inst/O	
		net (fo=1, routed)	0.739
1.143		Clock_IBUF	
	BUFGCTRL_X0Y0	BUFG (Prop_bufg_I_O)	0.030
1.173	r	Clock_IBUF_BUFG_inst/O	
		net (fo=230, routed)	0.739
1.912		Clock_IBUF_BUFG	
	SLICE_X9Y78	FDRE	
	r	Cycle_Count_reg[0]/C	
Slack:		inf	
Source:		Reset	
		(input port)	

Destination: Cycle_Count_reg[13]/R
 (rising edge-triggered cell FDRE
 clocked by Clock {rise@0.000ns fall@5.000ns period=10.000ns})
 Path Group: (none)
 Path Type: Hold (Min at Fast Process Corner)
 Data Path Delay: 0.890ns (logic 0.144ns (16.192%) route
 0.746ns (83.808%))
 Logic Levels: 1 (IBUF=1)
 Clock Path Skew: 1.912ns (DCD - SCD - CPR)
 Destination Clock Delay (DCD): 1.912ns
 Source Clock Delay (SCD): 0.000ns
 Clock Pessimism Removal (CPR): -0.000ns

Location	Delay type	Incr(ns)
Path(ns)	Netlist Resource(s)	

J26		0.000
0.000 r Reset (IN)		
	net (fo=0)	0.000
0.000 Reset		
J26	IBUF (Prop_ibuf_I_O)	0.144
0.144 r Reset_IBUF_inst/O		
	net (fo=42, routed)	0.746
0.890 Reset_IBUF		
SLICE_X9Y78	FDRE	
r Cycle_Count_reg[13]/R		

	(clock Clock rise edge)	0.000
0.000 r		
D23		0.000
0.000 r Clock (IN)		
	net (fo=0)	0.000

0.000		Clock		
	D23		IBUF (Prop_ibuf_I_O)	0.404
0.404	r	Clock_IBUF_inst/O		
			net (fo=1, routed)	0.739
1.143		Clock_IBUF		
		BUFGCTRL_X0Y0	BUFG (Prop_bufg_I_O)	0.030
1.173	r	Clock_IBUF_BUFG_inst/O		
			net (fo=230, routed)	0.739
1.912		Clock_IBUF_BUFG		
		SLICE_X9Y78	FDRE	
	r	Cycle_Count_reg[13]/C		